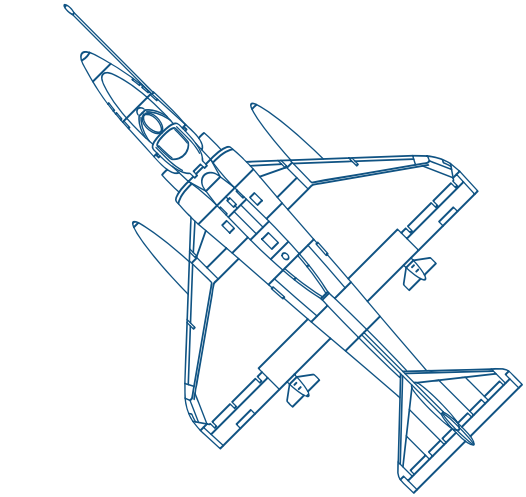
A-4E-C



A-4E COMMUNITY

for DCS World

Kneeboard Manual

Version 2.3

Revised 2025-06-04.

This manual is provided as a PDF and in the in-game kneeboard. - "RSHIFT+K"

Not for real world flight.

Developers

Gospadin, Heclak, JNelson, plusnine, Farlander, gyrovague, kryb/Archimaede, Merker, Jones, Nero

Join the community!

Thanks to the community for their many contributions, and to you, for flying the A-4E-C for DCS World. Visit the project page to download the latest version, report bugs, or to get involved on Discord:

https://community-a-4e.github.io/

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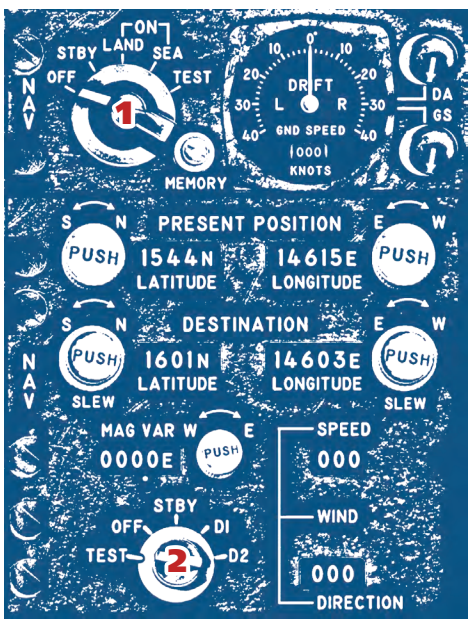
6 - Orientation
Throttle, Stick, Instrument Panel, Console, Controls Indicator, Gamepads

Flight Plan Waypoints

	LOCATION	LATITUDE	LONGITUDE
00			
01			
02			
03			
04			
05			
06			
07			
08			
09			
10			

AN/APN-153 DOPPLER NAVIGATION RADAR - 6-4, 2.

Set the Mode Selector **(1)** to STBY at start-up. In flight, set the mode to LAND or SEA as required.



AN/ASN-41 NAVIGATION COMPUTER - 6-4, 3.

Set the Function Selector **(2)** to STBY at start-up. Proceed to input magnetic variance, wind direction and speed. While in the STBY position, Present Position can be set, and the destination push-to-set knobs input to D1. In the D1 or D2 position, only the slew knobs provide input. For course bearing, set the BDHI switch to NAV CMPTR - 6-2, 29.

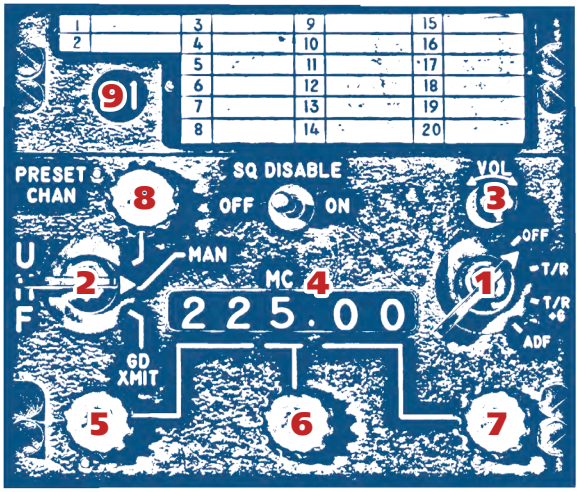
> Long flights or heavy maneuvers will cause the system to drift.

The first two waypoints from the Mission Editor are pre-set (D1 = WP1, D2 = WP0)

Channel Presets

CHN	FREQ
01	_____ . _____
02	_____ . _____
03	_____ . _____
04	_____ . _____
05	_____ . _____
06	_____ . _____
07	_____ . _____
08	_____ . _____
09	_____ . _____
10	_____ . _____
11	_____ . _____
12	_____ . _____
13	_____ . _____
14	_____ . _____
15	_____ . _____
16	_____ . _____
17	_____ . _____
18	_____ . _____
19	_____ . _____
20	_____ . _____

- Channel presets are defined in the Mission Editor.
- Radio Microphone Push to Talk (PTT) - "RALT+\" is located on the throttle handle, 6-1, 2.
- Additional PTT bindings for VOIP are available, including a Guard transmit when in TR+G mode.



AN/ARC-51A UHF Radio - 6-4, 5.

1. UHF Radio Mode . . . . . TR+G
2. Frequency Mode Switch . AS REQ'D
3. Volume Knob . . . . . AS REQ'D

FREQUENCY MODES

MAN: The frequency is displayed (4), and manually set from 225 - 399 MHz, using 10 MHz (5), 1 MHz (6), and 50 kHz (7) knobs.

PRESET CHAN: Set using Preset Channel Selector (8), and displayed (9).

GD XMIT: Sets frequency to 243 MHz (Guard).

UHF RADIO MODES

OFF: The radio is powered off.

T/R: Transmits and recieves on a frequency set by the Frequency Mode switch.

T/R+G: Also listens on 243 MHz AM (Guard).

ADF: Automatic Direction Finding - 1-3

Keep the canopy OPEN to talk with the ground crew, including refueling, rearming, or other procedures!

AN/ARN-25 Automatic Direction Finding (ADF) Equipment - 1-2

Switches UHF from the blade antenna to the ADF antenna, which rotates to the null point. Relative bearing is displayed on BDHI Needle 1.

1. **UHF Radio Mode. . . . . ADF** - 6-4, 5.
2. **BDHI Needle 1. . . . . CHECK** - 6-2, 2.

ADF mode garbles UHF transmission and reception.

AN/ARN-52 TACAN

With a line of sight to a source, TACAN provides continuous directional information on BDHI Needle 2, and distance information (to 300 miles) in the range window of the BDHI instrument.



1. **Mode . . . . . REC** - 6-4, 7.
2. **Channel Selector Switch 10s**
3. **Channel Selector Switch 1s**
4. **Volume . . . . . AS REQ'D** - for Morse audio ident
5. **BDHI Switch. . . . . TACAN** - 6-2, 29.
6. **BDHI Needle 2. . . . . CHECK** - 6-2, 2.

- Set the mode to A/A to recieve TACAN from airborne signals like tankers.
- X-band signals have priority over Y-band signals with the same number.

AN/ARA-63 MCL (ICLS) - 6-4, 11.

1. **MCL Power Switch . . . . . BIT** - Hold for system test
2. **MCL Power Switch . . . . . ON**
3. **MCL Channel Selector Knob . . AS REQ'D**
4. **Vert./Horiz. Guides . . . . . check ADI** - 6-2, 8.

Match the indicated glide slope and heading.

Airfield Channels & Frequencies

CHN	CODE	NAME	FREQ	BRG
01				
02				
03				
04				
05				
06				
07				
08				
09				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				
21				
22				
23				
24				
25				
26				
27				
28				
29				
30				

If the ground crew performs a rearm, the information on this page is updated.

- Night Vision Goggles (NVG) equipment can be added in the Mission Editor.



Fuel Stores

INT: _____ LBS.

EXT: _____ LBS.

Station Loadouts - AGM-45 Seeker - "LSHIFT+1, 2, 4, 5"

		QTY.
1.	_____	____
2.	_____	____
3.	_____	____
4.	_____	____
5.	_____	____

AN/ALE-29A Programmer - Countermeasures, 3-9

You can set these options in the Mission Editor, or while the engine is OFF, you can use the noted bindings to request the ground crew configure CBU and countermeasures programming.

CBU CONFIGURATION

CBU 1/A	2	Tubes Per Pulse	
CBU 2/A	_____	Tubes Per Pulse	- "RSHIFT+RALT+2"
CBU 2B/A	_____	Tubes Per Pulse	- "RSHIFT+RALT+3"

COUNTERMEASURES

BURST	_____	Bursts	- "RSHIFT+RALT+4"
BURST INTVL	_____	Seconds	- "RSHIFT+RALT+5"
SALVOS	_____	Tubes Per Pulse	- "RSHIFT+RALT+6"
SALVOS INTVL	_____	Seconds	- "RSHIFT+RALT+7"

Stations & Quantities

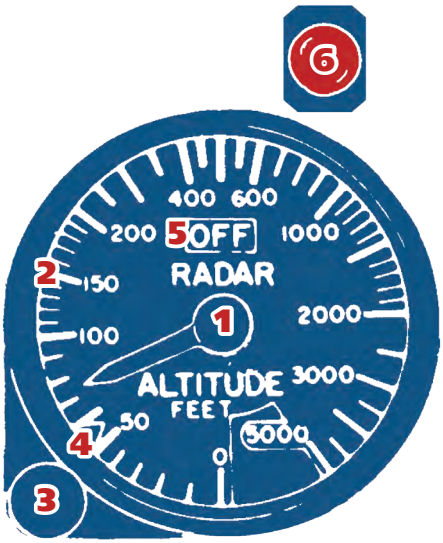
			STATIONS 1 & 5	STATIONS 2 & 4	STATIONS 3
A/A	AIM-9	B, J, P, P3, P5	❶	❶	
A/G	AGM-45	A, B	❶	❶	
ROCKETS	LAU-3	19 × FFAR, HYDRA	❶	❶❷	❶❷❸
	LAU-10	4 × ZUNI	❶	❶❷	❶❷❸
	LAU-68	7 × FFAR, HYDRA	❶	❶❷	❶❷❸
BOMBS	AN	M30A1	❶	❶	❶
		M57	❶	❶	❶❸❻
		M64	❶	❶	❶
		M65		❶	❶
		M66A2			❶
		M81, M88	❶	❶❺	❶❻
	BDU	33, 50LD, 50HD	❶	❺	❻
	M117			❶	❶
	MK	20	❶	❶❷	❶❷❸
		77 MOD0		❶	❶
		77 MOD1	❶	❶	❶❷
		81, 81 SE	❶	❶❺	❶❻
		82, 82 SE	❶	❶❷	❶❸❹❻
		83	❶	❶	❶❷❸
		84		❶	❶
	SUU	25 (LUU-2)	❶	❶	❶
PODS	CBU	1/A, 2/A, 2B/A		❶❷	
	HIPEG	MK4	❶	❶	❶
	ACMI	AN/ASQ-T50 TCTS	❶	❶	❶
	SMOKEWINDER		❶	❶	❶
FUEL TANKS	DFT	150 GAL	❶	❶	❶
		300 GAL		❶	❶
		400 GAL			❶

- Master Arm, Jettison, AWRS, 3-1
- Bombs, Bombing Computer, 3-2
- Rockets, 3-3
- Guns, HIPEG Gunpods, 3-4, 3-5
- AIM-9 (Sidewinder), 3-5
- AGM-45 (Shrike), 1-5, 3-6, 3-7
- Countermeasures, RHWS (RWR/ECM), 3-9
- CBU & Countermeasures Configuration, 1-5
- Fuel Tanks & Air Refueling, 4-4

AN/APN-141 Radar Altimeter

This pulse radar provides instantaneous altitude information to the pilot from 0 to 5000' terrain clearance above ground level (AGL), which is displayed on the indicator **(1)**. - 5-2, 26.

The indicator scale **(2)** displays 10' increments from 0' to 200', 50' increments from 200' to 600', 100' increments from 600' to 2000', and 500' increments from 2000' to 5000'.



A control knob **(3)** on the indicator controls power to the system. Turning the knob adjusts the altitude setting of the low-limit indexer **(4)**.

The radar altimeter operates in up to 50° of climb or dive, and up to 30° angles of bank, right or left.

The OFF flag **(5)** on the indicator appears when signal strength becomes inadequate, when power is lost, or when the system is turned off.

Low Altitude Warning System (LAWS)

The warning system warns the pilot of impending danger due to low altitude, and consists of two warning lights: the radar low limit warning light **(6)**, and the LAWS light on the Dashboard Indicator Warning Lights. - 6-2, 16.

If the indicator **(1)** drops below the low-limit indexer **(4)**, both warning lights come on, and an aural warning tone is activated for 2 seconds, alternating 700- to 1700-CPS tones with a repetition rate of 2 CPS. - Cycles Per Second (CPS = Hz)

AN/APG-53A Radar

The radar provides modes for SEARCH and mapping, two modes for TERRAIN CLEARANCE and obstacle avoidance (PLAN and PROFILE), and an AIR-TO-GROUND slant range mode for weapons delivery. Warm up the AN/APG-53A equipment at start-up.

The AN/APG-53A Control Panel houses the Radar Mode Switch, AoA Compensation Switch, the Antenna Tilt Switch and a control knob for the Obstacle Tone Volume. - 6-3, 10.

The Misellaneous Switches Panel houses the Radar Terrain Clearance Switch and Radar Range Switch. - 6-2, 29.

Additonal controls are located on the Radar Indicator. - 6-2, 20.

Radar Indicator

A set of range-elevation-azimuth reference lines **(1)** are printed on the scope. The scale and use of these lines varies in accordance with the radar mode in use.

Two flag indicators enable when the mode orients the display to a plan-view aircraft **(2)**, or a profile-view aircraft **(3)**.

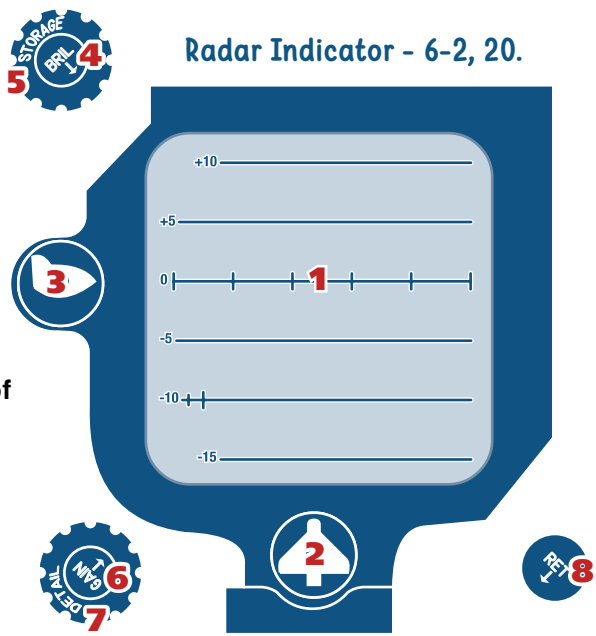
The Brilliance Knob **(4)** determines the brightness of the display, while the Storage Knob **(5)** determines how long the picture lasts.

The Gain Knob **(6)** varies the amplitude of the signals (sensitivity). This knob is usually three-quarters of its clockwise travel.

The Detail Knob **(7)** adjusts the vertical beam width in the Terrain Clearance Plan mode.

For night flying, the Reticle Knob **(8)** controls illumination of the range-elevation-azimuth reference lines. Seting the red Radar Indicator Filter Plate in the down position, changes the green display to red, with a brightness consistent with cockpit illumination.

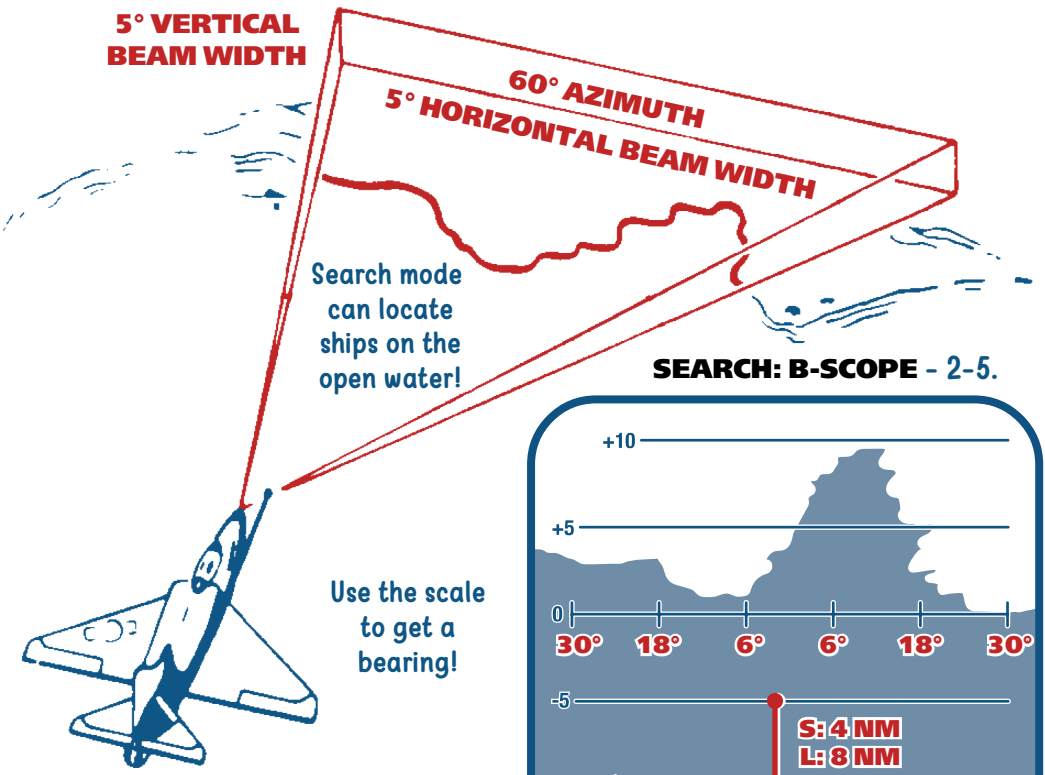
Radar Indicator - 6-2, 20.



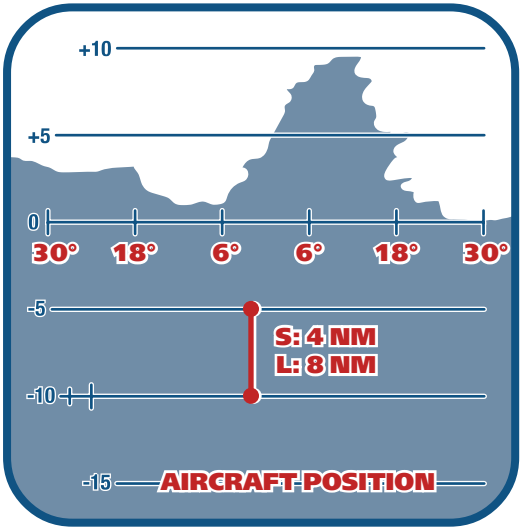
Search Mode

1. Radar AoA Compensation Switch . . . . . ON - 6-3, 10.
2. AN/APG-53A Radar Mode Selector . . . . . SRCH
3. Radar Antenna Tilt . . . . . AS REQ'D
3. Radar Range Switch . . . . . SHORT/LONG - 6-2, 29.
4. Radar Indicator Knobs . . . . . AS REQ'D - 6-2, 20.

To acquire a SEARCH picture, "snow" the scope by turning the Radar Indicator Gain knob to maximum, then reduce it slowly, until only a trace of snow remains. Adjust the Radar Antenna Tilt to select which terrain to scan.



SEARCH: B-SCOPE - 2-5.



If you experience performance issues in the cockpit, try enabling the AN/AGP-53A Radar Legacy Mode in DCS World's Special options menu for the A-4E-C. In Legacy Mode, the radar generates a significantly less detailed picture.

RANGE
S: 20 MILES
L: 40 MILES



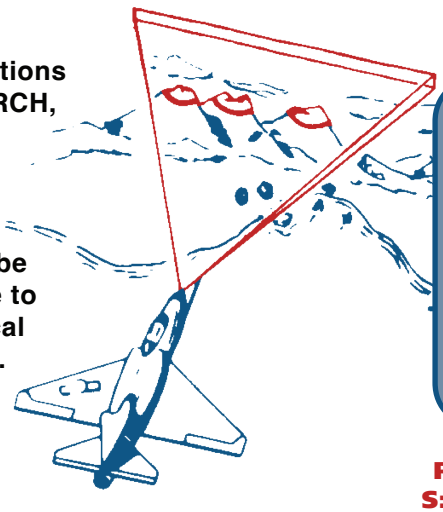
Terrain Clearance Mode

1. Radar AoA Compensation Switch . . . . . ON - 6-3, 10.
2. AN/APG-53A Radar Mode Selector . . . . . TC
3. Radar Antenna Tilt . . . . . AS REQ'D
4. Radar Terrain Clearance Switch . . PLAN/PROFILE - 6-2, 29.
5. Radar Range Switch . . . . . SHORT/LONG
6. Radar Indicator Knobs . . . . . AS REQ'D - 6-2, 20.
7. OBST Dashboard Warning Indicator . . . . . CHECK - 6-2, 16.
8. Radar Obstacle Tone Voume . . . . . AS REQ'D - 6-3, 10.

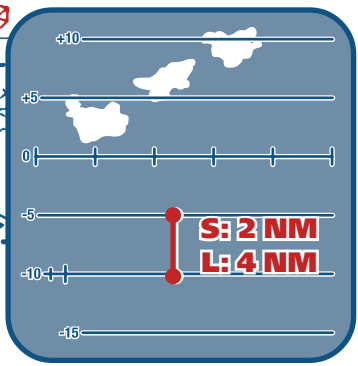
Plan

PLAN mode functions identical to SEARCH, except the range is halved.

In this mode, the Detail Knob can be turned clockwise to narrow the vertical beam width to 1°.



PLAN: B-SCOPE - 2-5.



RANGE
S: 10 NM
L: 20 NM



Profile

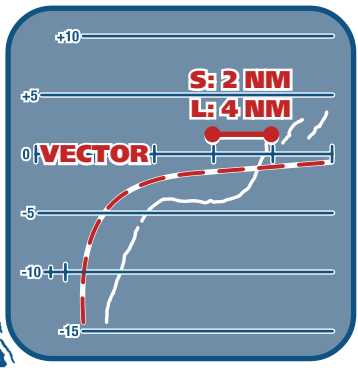
In PROFILE mode, obstacles are scribed on the indicator in relation to your vector. An imaginary reference line 1000 feet below the aircraft is also drawn, accounting for scope distortion.

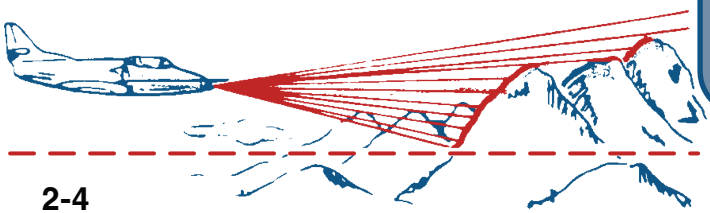
A warning tone is played when obstacles are within 1000 feet of the flight path.

PROFILE: E-SCOPE - 2-5.

RANGE
S: 10 NM
L: 20 NM

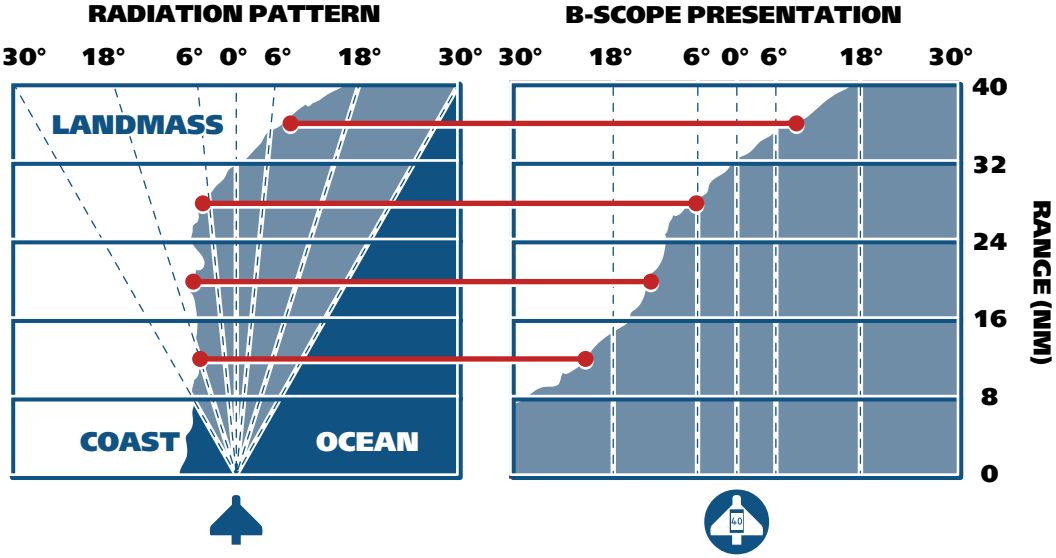




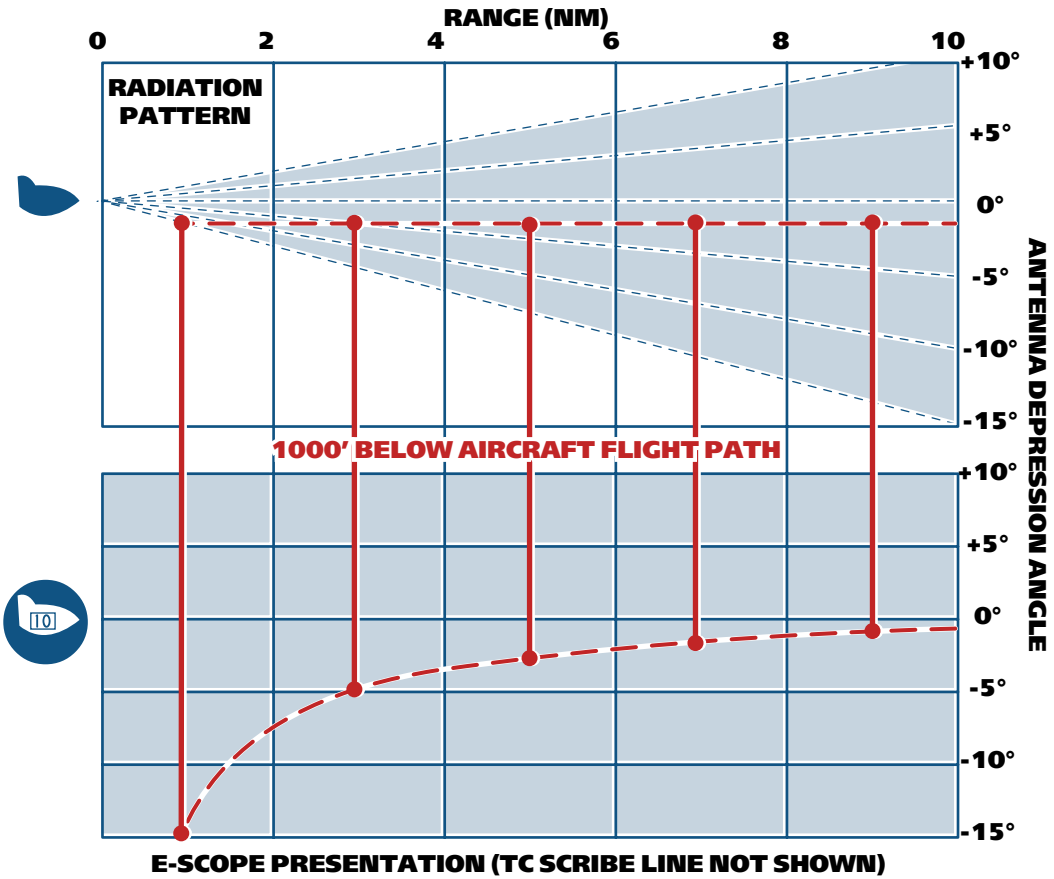


1000' BELOW AIRCRAFT
(DISPLAYED ON SCOPE)

B-Scope Distortion (Search & Plan)



E-Scope Distortion (Profile)



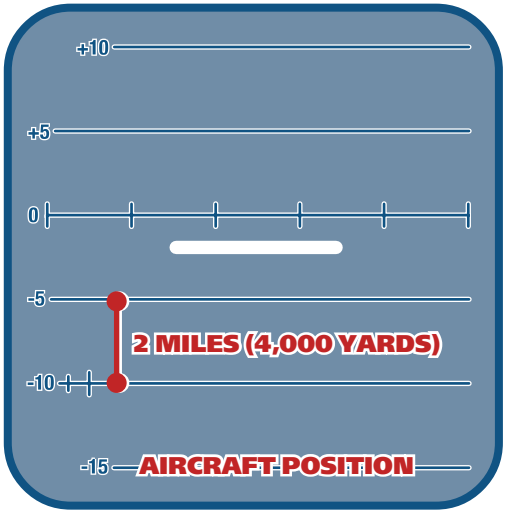
Air-to-Ground Mode

1. AN/APG-53A Radar Mode Selector . . . . . A/G - 6-3, 10.
2. Radar Indicator Knobs . . . . . AS REQ'D - 6-2, 20.
3. Gunsight Elevation . . . . . 0 MILS - 6-2, 14.
4. Initiate Dive . . . . . 10° OR MORE

In a A/G mode, while in a dive of 10° or more, and with the optical sight intercepting the ground, the horizontal bar on the indictor shows the slant range between the nose and the ground, to a maximum of 7.5 nautical miles (15,000 yards).

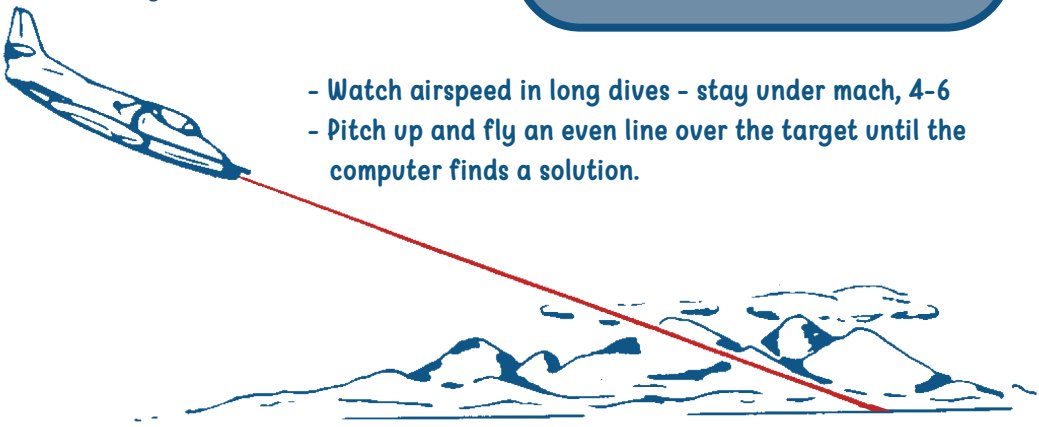
The antenna is fixed at 0°, and it sweeps vertically until it makes contact with the ground, at which time the indicator bar shows the distance to the ground.

This mode is also used to inform the CP-741/A - 3-2 Bombing Computer for automatic weapons release.



COMPUTER RELEASE TIPS

- Ensure the aircraft is properly trimmed out, 6-1
- Roll in toward target and keep dive 30° or more at sufficient height AGL
- Watch airspeed in long dives - stay under mach, 4-6
- Pitch up and fly an even line over the target until the computer finds a solution.



Master Arm Procedure

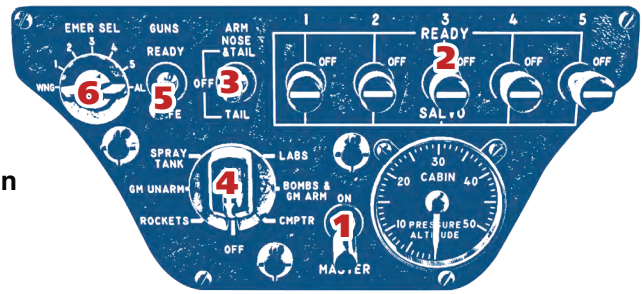
ARMAMENT PANEL - 5-2, 34.

1. Master Armament Switch . . . . . ON - "0"
2. Station Selector Switches . . . READY AS REQ'D - "1, 2, 3, 4, 5"
3. Bomb Arm Switch . . . . . NOSE & TAIL - "6"
4. Weapon Function Selector Switch . . . AS REQ'D - "7, 8"
5. Guns Charging Switch . . . . . AS REQ'D - "9"

The MASTER ARMAMENT SWITCH must be in the ON position to fire guns or release weapons. Weapon firing systems require generator power, and are disabled by deploying landing gear.

Jettison

Set the Emergency Release Selector Switch (6) to the desired station. The WNG position selects stations 1, 2, 4 and 5. The AL position selects all stations. To jettison selected stations, pull the Emergency Stores Release Handle. - "LCTRL+J", 6-2, 31.



AWE-1 Aircraft Weapons Release

AIRCRAFT WEAPONS RELEASE SYSTEM PANEL - 6-2, 35.

1. Quantity Selector . . . . . AS REQ'D
2. Drop Interval . . . . . AS REQ'D - under 80ms unsafe!
3. AWRS Mode . . . . . AS REQ'D - Mult. Sw. = 10x interval

Configure the armament panel as required. - 3-1, 3-4

Use SALVO mode for simultaneous release of all munitions across all armed stations.



For single-munition releases, set RIPPLE-SINGLE.

In PAIR modes, both stations in the pair must be set to READY.
Stations 1 and 5 are paired.
Stations 2 and 4 are paired.
Station 3 is not paired.



Station Pairing Diagram

Low-Drag (Slick) Bombs

WEAPON	DIVE ANGLE	REL HGT AGL	SPEED KIAS	RETICLE
LO-DRAG	30°	1000'	300	100
LO-DRAG	20°	2000'	450	50
LO-DRAG	30°	3000'	450	40
LO-DRAG	45°	4000'	400	40
	└ 6-2, 8.	└ 6-2, 26.	└ 6-2, 9.	└ 6-2, 14.

CP-741/A Computer Release (CCRP)

The computer calculates the release point for low-drag bombs, releasing weapons in accordance with AWE-1 settings. In a dive, sight the target and ensure a lock on the AN/APG-53A Radar Indicator. Press and hold the Bomb Release Button over the target. The LABS Dashboard Warning Indicator Light turns on, and the weapon release tone is heard. In STEP mode, the tone and light clear when the Bomb Release Button is released. In RIPPLE, the tone and light clear once the release sequence completes. - Computer Release Tips, 2-6

1. AN/APN-141 Radar Altimeter . . . . . ON - 2-1
2. AN/APG-53A Radar Mode . . . . . A/G - 2-2, 2-6.
3. Gunsight Elevation . . . . . 0 MILS - 6-2, 14.
4. Weapon Function Selector . . . . . CMPTR - 3-1, 5.
5. Master Armament Switch . . . . . ON - 3-1, 1.
6. Station Selector Switches . . . . . AS REQ'D - 3-1, 3.
7. Bomb Release Button . . DEPRESS/COMMENCE - 6-1, 9.

The computer does not account for wind or yaw.

MK-82 Snakeyes

BASELINE

1. Gunsight Elevation . . . . . 160 MILS - 6-2, 14.
2. Speed . . . . . 400 KIAS - 6-2, 9.
3. Release Height AGL . . . . . 400' AGL - 6-2, 26.

ADJUSTMENT

4. Speed . . . . . -100 KIAS = +50 MILS
5. Release Height AGL . . . . . +100' AGL = +50 MILS

Rocket Fire Modes

WEAPON FUNCTION SELECTOR KNOB - 3-1, 5.

1. ROCKETS

   A. Gun-Rocket Trigger . . Fire GUNS & ROCKETS - 6-1, 8.

2. GM UNARM

   A. Gun-Rocket Trigger . . . . . Fire GUNS - 6-1, 8.

   B. Bomb Release Button . . . . . Fire ROCKETS - 6-1, 9.

LAU-3 / LAU-68 FFAR, HYDRA

WEAPON	DIVE ANGLE	HGT AGL	SPEED KIAS	RETICLE
FFAR MK 1	20°	5000’	375	40
FFAR MK 1	20°	4000’	400	30
FFAR MK 1	20°	3000’	400	20
FFAR MK 1	20°	2500’	400	10
FFAR MK 1	20°	2000’	400	0
FFAR MK 1	30°	2500’	400	0
FFAR MK 1	45°	3500’	400	0

L 6-2, 8. L 6-2, 26. L 6-2, 9. L 6-2, 14.

LAU-10 / ZUNI

WEAPON	DIVE ANGLE	HGT AGL	SPEED KIAS	RETICLE
ZUNI	20°	5500’	350	10
ZUNI	20°	4500’	375	5
ZUNI	20°	3500’	380	0
ZUNI	30°	5000’	380	0
ZUNI	45°	7000’	380	0

L 6-2, 8. L 6-2, 26. L 6-2, 9. L 6-2, 14.

MK 12 Cannons - 2 * 100 rounds, 1000 rpm.

ARMAMENT PANEL - 5-2, 34.

1. Guns Charging Switch . . . . . READY - "9"

> When the GUNS CHARGING SWITCH is set to the ON position, an amount of nitrogen charge is expended. Once expended entirely, the ground crew must rearm the aircraft to maintain the guns.
> SAFE ALL GUNS BEFORE LANDING.

Charging and clearing is audible - if you don't hear it, you're out of nitrogen!

The MASTER ARMAMENT SWITCH must be in the ON position to fire guns, see 3-1

MK 4 HIPEG Gunpod - 750 rounds, 1000 rpm.

GUNPOD PANEL - 6-3, 2.

1. LH/CTR/RH Gunpod Switch . . READY AS REQ'D
2. Gunpod Switch . . . . . CHARGE

> When the GUNPOD SWITCH is set to the CHARGE position, gunpods with stations in the READY position gunpods are immediately charged. Keep the switch in the CHARGE position to fire gunpods.
>
> When the GUNPOD SWITCH is set to the CLEAR position, all gun pods on the aircraft will be cleared, rendering them inoperable until the ground crew rearms the aircraft and maintains the gunpods.
> CLEAR ALL GUNPODS BEFORE LANDING.

LH = Station 2 CTR = Station 3 RH = Station 4

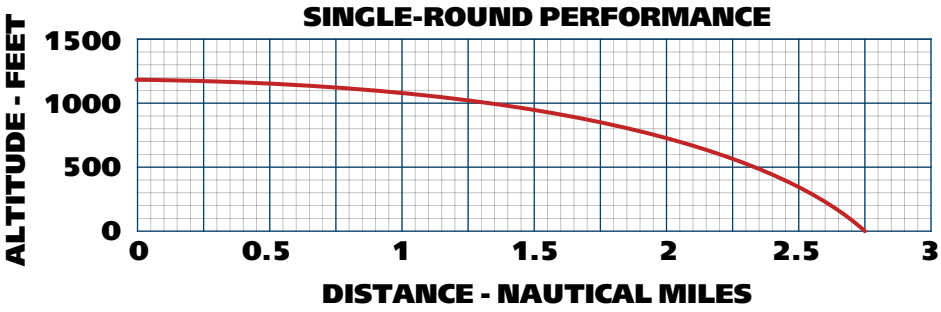
Ready gunpods on the GUNPOD PANEL, not the ARMAMENT PANEL!

Gun Strafing Sight Angles

WEAPON	REL ANGLE	HGT AGL	SPEED KIAS	RETICLE
MK12/MK4	-10°	1000'	400	0
MK12/MK4	-10°	2000'	300	27
MK12/MK4	-10°	2000'	450	17
MK12/MK4	-15°	2000'	450	15
MK12/MK4	-20°	1500'	450	0

L 6-2, 26. L 6-2, 9. L 6-2, 14.

Gun Performance



AIM-9 (Sidewinder)

ARMAMENT PANEL - 6-2, 34.

1. Weapon Function Selector . . BOMBS & GM ARM - 3-1, 5.
2. Station Selector Switches . . . READY AS REQ'D - 3-1, 3.

MISCELLANEOUS SWITCHES PANEL

3. Shrike/Sidewinder Volume Knob . . . . AS REQ'D - 6-2, 29.

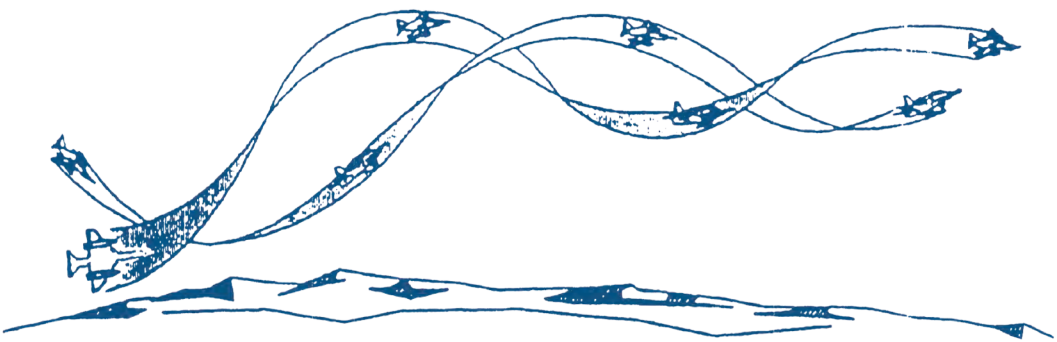
DELIVERY

FOX TWO . . . . . GUN-ROCKET-TRIGGER - 6-1, 8.
or BOMB RELEASE BUTTON - 6-1, 9.

- Sidewinder lock tone behavior is limited to rear-aspect only. However, missiles track targets normally after launch.

DOGFIGHTING - LOSE SIGHT, LOSE THE FIGHT

- Slow enemy down, bleed their energy
- Don't climb above 15,000 feet
- 90% RPM, nose up, rudder to turn
- Maintain speeds of 150-170 KIAS
- Outmaneuver in horizontal scissors



AGM-45 (Shrike) - B has a more powerful rocket motor than A model

PRE-TAXI - 5/2

1. Mission Editor . . . . . PROCEED to 4.
2. Request Rearming . . . . . AS REQ'D
3. Confirm Seeker Head RF Kit . . . KNEEBOARD - "RShift+K", 1-5.

DIRECT: Launch unlocks fins
LOFT: Fins unlock < 18,000' (not recommended)

ARMAMENT PANEL - 6-2, 34.

4. Weapon Function Selector . . BOMBS & GM ARM - 3-1, 5.
5. Station Selector Switches . . . READY AS REQ'D - 3-1, 3.

MISCELLANEOUS SWITCHES PANEL

6. Shrike/Sidewinder Volume Knob . . . . AS REQ'D - 6-2, 29.

DELIVERY

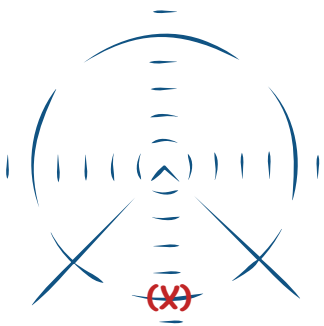
SHOTGUN . . . . . BOMB RELEASE BUTTON - 6-1, 9.

Armed AGM-45 stations seek and fire in order: 1, 5, 2, 4 - 3-1

ELEVATED ATTACK
- Approach target from > 20,000' AGL
- 60° nose up on target
- SHOTGUN at > 22,000' AGL
- Egress perpendicular to target

DIVE ATTACK
- Set gunsight to 0 mils
- Approach target from > 8,000' AGL
- At < 10 nm to target, dive
- SHOTGUN on target at -50 mils (X)

POP-UP ATTACK
- At 25 nm to target, dive to <500' AGL
- At < 16 nm, 60° nose up on target
- SHOTGUN at > 5000' AGL
- Return to < 500' AGL
- Egress



- Dynamic spawns assign MK 50 RF Kit.
- Confirm RF Kit loadout in kneeboard after rearming. - "RShift+K", 1-5.
- Mismatched loadouts produce unreliable tracking and seeking results.

AGM-45 RF Kits vs. Ground Units

R: Rangefinder S: Search ST: Search and Tracking T: Tracking
B: Big Bird C: Clam Shell D: Dog Ear F: Flat Face TS: Tin Shield
Items in red don't provide tone or needles, but the missile will track the radar.

RF KIT	SA-2	SA-3	SA-5	SA-6	SA-8	SA-10	SA-11	SA-13	SA-15	SA-19	EWR
MK 22	T/R	–	T	–	–	–	–	T	T	–	D
MK 23	T/R	–	TS/T	–	–	TS/B/C	S	T	S	S	–
MK24.5	T/R	–	TS/T	–	–	T	–	T	–	S	–
MK24.34	T/R	–	TS/T	–	–	TS/B/C	–	T	–	S	–
MK 25	T/R	–	T	–	–	–	–	T	T	–	D
MK 36	T/R	T	T	T	–	T	T	T	–	–	–
MK 37	F/R	F	F/T	S	S	–	–	T	–	–	–
MK 49.0	T/R	T	T	T	–	T	T	T	–	–	–
MK 49.1	T/R	T	T	T	–	T	T	T	–	–	–
MK 50	T/R	–	TS/T	–	–	TS/B/C	–	T	S/T	S	D

RF KIT	CH-SA-4	HAWK	NASAMS	PATRIOT	RAPIER	ROLAND	FIRECAN	GEPARD
MK 22	T	T	–	ST	S	–	T	–
MK 23	S/T	T	–	–	S	–	T	S
MK24.5	S	T	–	–	S	–	T	S
MK24.34	S	T	–	–	S	–	T	–
MK 25	S	T	–	ST	S	–	T	–
MK 36	S	T	S	–	S	T	T	–
MK 37	S	S/T	–	–	S	–	T	–
MK 49.0	S	T	–	–	S	–	T	–
MK 49.1	S	T	–	–	S	–	T	–
MK 50	S/T	T	–	ST	S	–	T	S

- Dynamic spawns assign MK 50 RF Kit.
- Confirm RF Kit loadout in kneeboard after rearming. - "RShift+K", 1-5.
- Mismatched loadouts produce unreliable tracking and seeking results.

AGM-45 RF Kits vs. Naval Units

S: Search ST: Search and Tracking T: Tracking

Items in red don't provide tone or needles, but the missile will track the radar.

RF KIT	GRISHA	KIROV	KRIVAK	MOLNIYA	NEUT.	T 052B	T 052C	T 054A
MK 22	T	T	T	–	T	T	–	T
MK 23	T	S/T	T	S/T	S/T	S/T	S/T	S/T
MK24.5	T	S/T	T	S/T	S/T	T	S/T	T
MK24.34	T	T	T	–	T	T	–	T
MK 25	T	T	T	–	T	T	–	T
MK 36	T	S/T	T	S/T	S/T	S/T	S/T	S/T
MK 37	T	T	T	–	T	T	–	T
MK 49.0	S/T	T	S/T	–	T	S	–	T
MK 49.1	S/T	T	S/T	–	T	S	–	S
MK 50	T	S/T	T	S/T	S/T	S/T	S/T	S/T

└ also SLAVA and KUNETSOV

RF KIT	LEANDER	DE MAYO	BURKE	NIMITZ	PERRY	TICON	–	–
MK 22	T	–	–	T	T	–	–	–
MK 23	T	S/T	–	T	T	–	–	–
MK24.5	T	–	–	T	T	–	–	–
MK24.34	T	–	–	T	T	–	–	–
MK 25	T	–	–	T	T	–	–	–
MK 36	T	–	T	T	T	T	–	–
MK 37	T	–	S/T	T	S/T	S/T	–	–
MK 49.0	T	–	–	T	T	–	–	–
MK 49.1	T	–	–	T	T	–	–	–
MK 50	T	S/T	–	T	T	–	–	–

- Dynamic spawns assign MK 50 RF Kit.
- Confirm RF Kit loadout in kneeboard after rearming. - "RShift+K", 1-5.
- Mismatched loadouts produce unreliable tracking and seeking results.

AN/ALE-29A Chaff Dispensing System

CHAFF PANEL - 6-2, 11.

1. Chaff Power Switch . . . . . PWR - "LCTRL+Q"
2. Dispenser Select . . . . . AS REQ'D
3. Supply program . . . . . AS REQ'D
   - A. JATO Firing Button . . . . BURST - "Q", 6-3, 4.
   - B. Chaff AUTO Pushbutton . . . . . SALVO - "LALT+Q"



With the engine OFF, the ground crew can set BURST and SALVO programs. - 1-4
Abort program early by setting the Chaff Power Switch to the OFF position.

AN/APR-25 Radar Homing and Warning System (RHWS) (RWR)

ECM PANEL - 6-2, 10.

1. APR-25 Switch . . . . . ON
2. Audio ALQ Switch APR-25
3. Function Selector . . . . . REC
4. APR-27 Switch . . . . . ON
5. APR-27 Test . . . . . PRESS - Bit Test.
6. PRF Volume . . . . . AS REQ'D
7. Missile Alert Volume . . AS REQ'D



With the APR-25 Switch **(1)** in the ON position, and the Function Selector **(3)** in the REC position, operation hum and signals from E-, G-, and I-band radars are audible at PRF Volume **(6)**. With the APR-27 Switch **(4)** in the ON position, the SAM light indicates an SA-2 launch, and an oscillating alert is audible at Missile Alert Volume **(7)**. With the Function Selector in RPT, electronic countermeasures will attempt to jam incoming radar locks.

DETECTED RADARS:
- AAA (SON-9 Fire Can)
- EWRs (ship, aircraft, ground)
- SAMs (SA-2, -3, -5, -6, -8, -10, -11, -13, and -19)
- SPAAA (Gepard, Vulcan, Shilka)
- Aircraft S/T radars

RPT mode's ECMs can give you away to other radars!

 - The ECM Panel can be removed in the Mission Editor.

Automatic Flight Control System

AUTOPILOT CONTROL PANEL (AFCS) - 6-3, 12.

1. STANDBY
   - A. AFCS Standby Switch . . . 90 SEC. WARM-UP - "LALT+A"
2. ENGAGE & ATTITUDE HOLD
   - A. Bank Angle . . . . . 5 - 70° - 6-2, 8.
   - B. Pitch Angle . . . . . under 60°
   - C. AFCS Engage Switch . . . . . ENGAGE - "A"

Lat./Long. attitude maintained at engage and disengage. Control Stick Steering (CSS) at 3 lbs. stick pressure.

3. STABILITY AUGMENTATION (Yaw Damper) - 5-3, 8.
   - A. Stability Augmentation Switch . . STAB AUG - "LSHIFT+S"

Additional modes require AFCS switch is engaged:

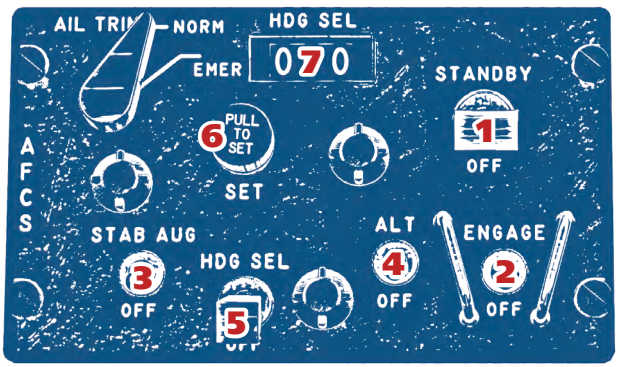
4. ALTITUDE HOLD - Careful, this switch can stick in engaged position!
   - A. Climb . . . . . 3500 - 4500' per MIN. - 6-2, 3.
   - B. Altitude Switch . . . . . ALT - "LSHIFT+A"
5. HEADING SELECT
   - A. Bank Angle . . . . . under 5° - 6-2, 8.
   - B. Pitch Angle . . . . . under 60°
   - C. Heading Select Switch . . . . . HDG SEL - "LCTRL+LSHIFT+A"

In Heading Select, the AFCS maintains a bank angle from 22° to 32°. When disengaged, the AFCS rolls to level.

6. HEADING SELECT PULL-TO-SET KNOB
7. HEADING SELECT DISPLAY

Disengage all modes and return the AFCS to standby by pressing Autopilot Override (AFCS Emergency Disconnect) on the Control Stick.

- "LCTRL+A", 6-1, 6.

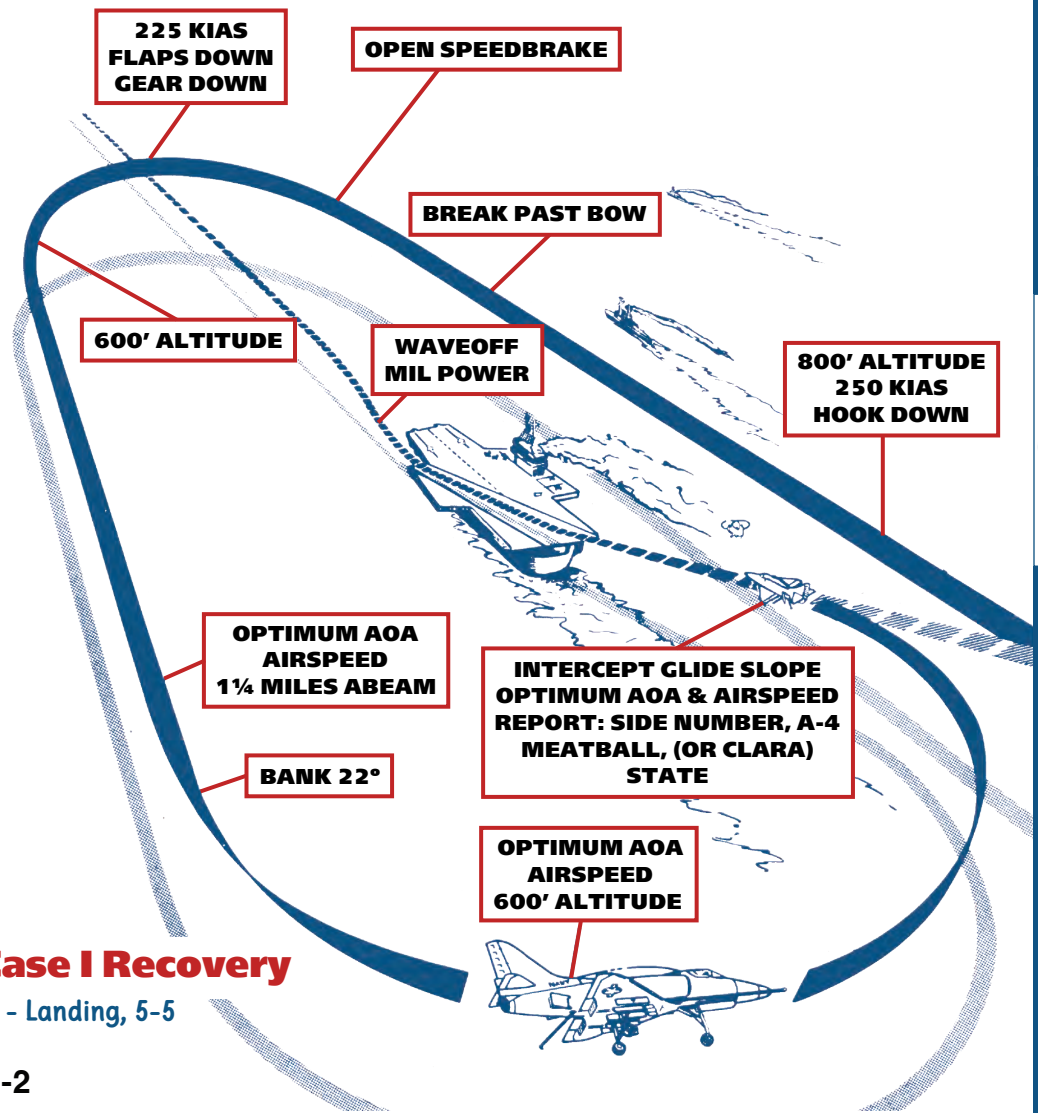


Approach Power Compensator (APC)

The APC controls throttle from 70% to MIL, and is designed to maintain the optimum AoA of 17.5 units, resulting in an optimum approach speed on the glide slope, and during normal maneuvers in the landing pattern at any landing gross weight.

APC CONTROL PANEL - 6-3, 3.

1. APC Power Switch . . STBY (15 SEC. WARM-UP)
2. APC Power Switch . . . . . ENGAGE
3. Throttle Position . . . . . 25 - 70% - 6-2, 15.
4. Allow system to take control of throttle - Hands OFF 15 sec.
5. Control Stick . . . . . OPTIMUM AoA - 6-2, 15.



Case I Recovery

- Landing, 5-5

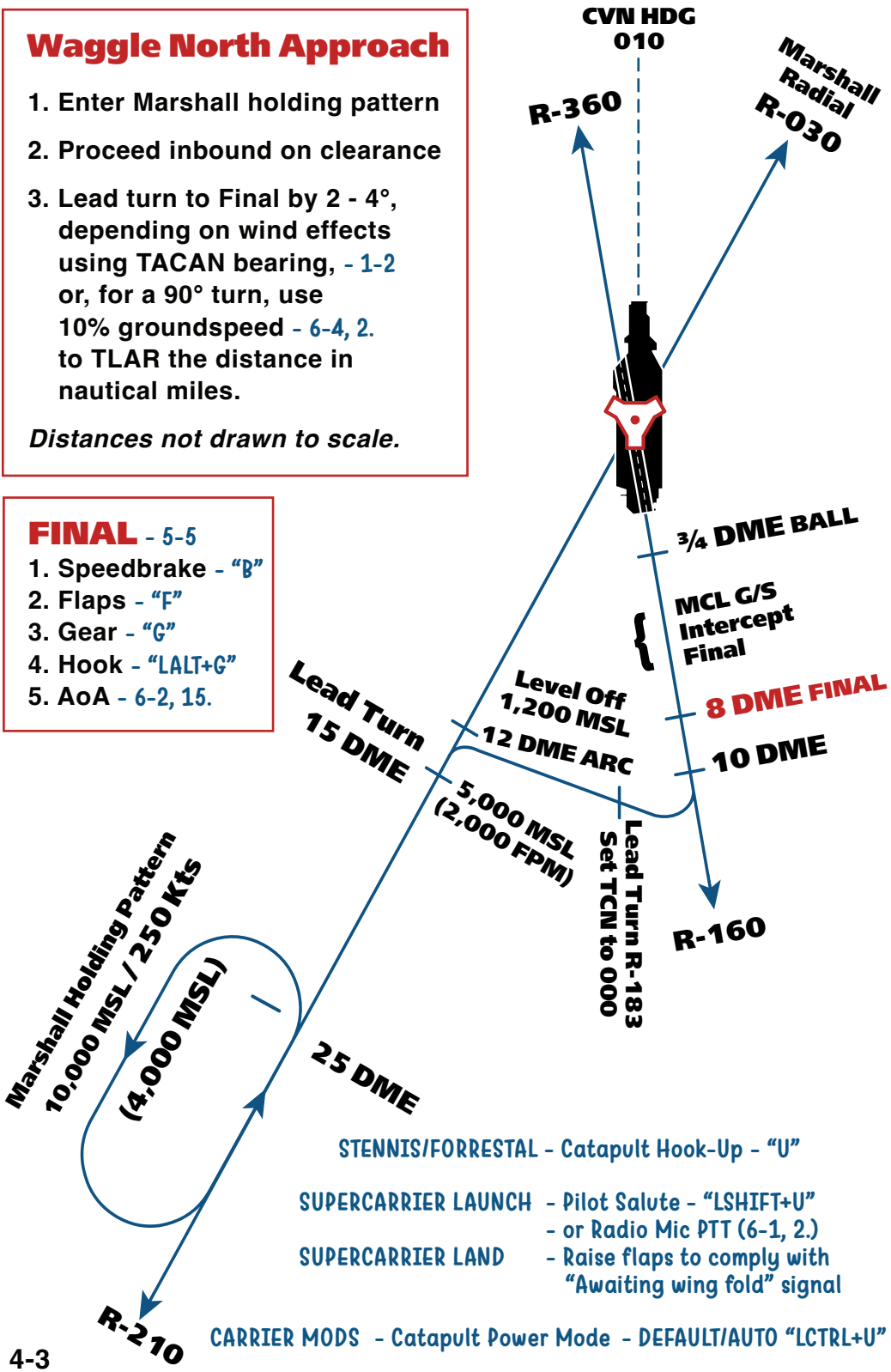
Waggle North Approach

1. Enter Marshall holding pattern
2. Proceed inbound on clearance
3. Lead turn to Final by 2 - 4°, depending on wind effects using TACAN bearing, - 1-2 or, for a 90° turn, use 10% groundspeed - 6-4, 2. to TLAR the distance in nautical miles.

Distances not drawn to scale.

FINAL - 5-5

1. Speedbrake - "B"
2. Flaps - "F"
3. Gear - "G"
4. Hook - "LALT+G"
5. AoA - 6-2, 15.



STENNIS/FORRESTAL - Catapult Hook-Up - "U"

SUPERCARRIER LAUNCH - Pilot Salute - "LSHIFT+U"
- or Radio Mic PTT (6-1, 2.)

SUPERCARRIER LAND - Raise flaps to comply with "Awaiting wing fold" signal

CARRIER MODS - Catapult Power Mode - DEFAULT/AUTO "LCTRL+U"

Fuel Quantity

INTERNAL FUEL

1. Fuselage Tank Capacity . . . . . 1612 LBS.
2. Wing Tank Capacity . . . . . 3830 LBS.
3. Fuel Quantity Indicator . . . . . CHECK - 6-2, 24.

EXTERAL FUEL STORES

4. Internal-External Fuel Switch . . . PRESS - 6-2, 29.
5. Fuel Quantity Indicator . . . . . CHECK - 6-2, 24.

Fuel Dumping

ENGINE CONTROL PANEL - 6-3, 8.

1. Fuel Dump Switch . . . . . DUMP - "LSHIFT+D"
2. Fuel Dump Rate . . . . . 680 LBS. PER MIN.



Air Refueling

NAVIGATE TO TANKER POSITION

1. TACAN Mode Switch . . . . . A/A - 6-3, 7.
2. TACAN Channel . . . . . AS REQ'D
3. BHDI Switch . . . . . TACAN - 6-2, 29.

TANKER CONTACT

4. Radio tanker . . . . . PRE-CONTACT - 1-2
5. Match tanker altitude . . . . AS DIRECTED - 6-2, 7.
6. Match tanker airspeed . . . AS DIRECTED - 6-2, 9.
7. Contact . . . . . TAKING FUEL

REFUEL EXTERNAL STORES

8. Drop Tank Pressurization Switch . . . . . FLIGHT REFUEL - "LCTRL+D" - 6-3, 8.

The Fuel Dump and Drop Tank Pressurization Switches are 3-position switches. The listed keyboard shortcuts cycle between the three positions.



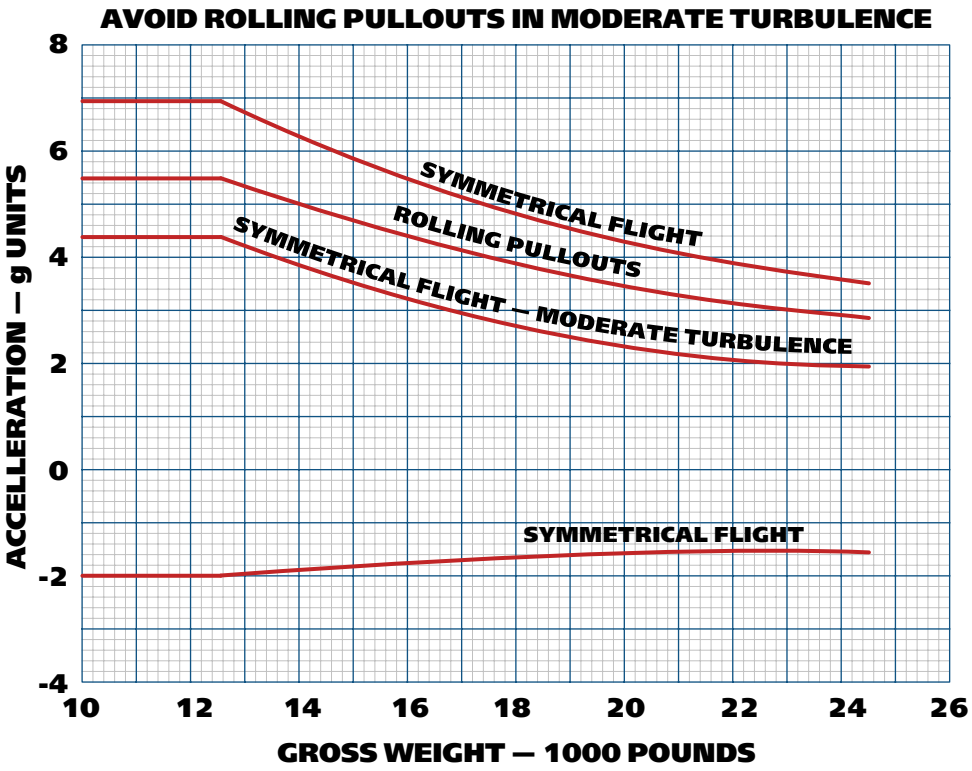
Maximum Gross Weight Limits

1. Field Takeoff . . . . . 24,500 LBS.
2. Field Landing (min. rate of descent) . 16,000 LBS.
3. Catapult . . . . . 24,500 LBS.
4. FCLP (or above min. rate of descent) . 14,500 LBS.

3° GLIDE SLOPE

GND SPEED	RATE of DESCENT
150	795
140	745
130	690
120	635
110	585

Structural Acceleration Limits



Station Limits and Thrust

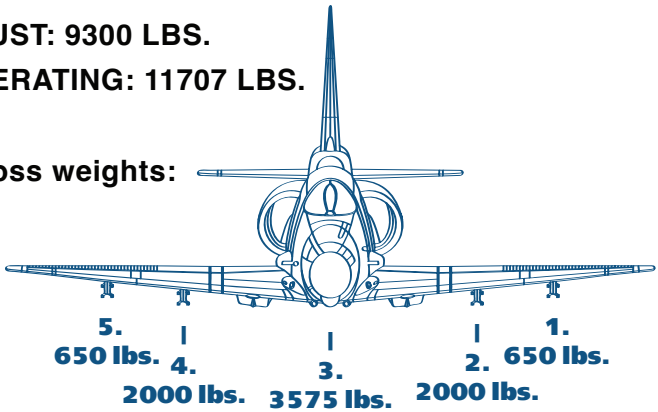
PW J52-P-8A MAX. THRUST: 9300 LBS.

EMPTY: 11587 LBS. OPERATING: 11707 LBS.

MAX: 24502 LBS.

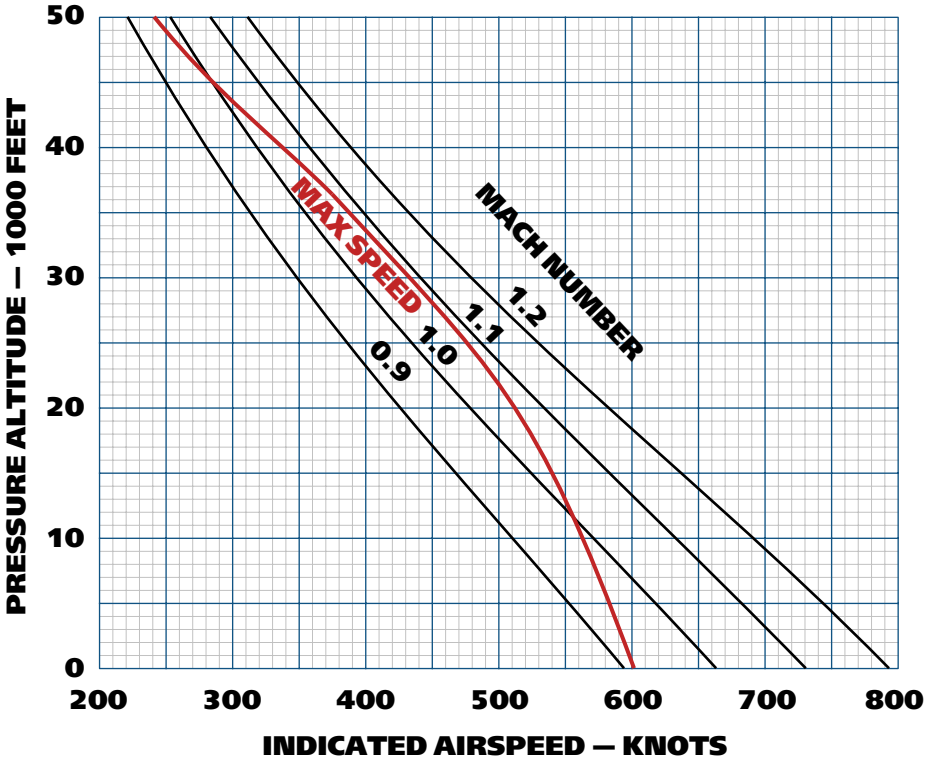
Weapon station max. gross weights:

Spool time can be quite long: keep RPMs high and use flight control surfaces for speed control!



Airspeed Limits

1. With no external stores and with landing gear, flaps and arresting hook retracted:



2. With 300-gallon drop tanks . . . . . 575 KIAS - 5-2, 9.
   or MACH 0.9 (whichever is lower)
3. With landing gear and/or flaps extended:
   A. Zero yaw . . . . . 225 KIAS
   B. Unrestricted yaw . . . . . 170 KIAS

STRAFING SPEED: 350-450 KIAS - 3-3
LANDING GEAR OVERSPEED DAMAGE: 185 KIAS (locks gear)
INVERTED FLIGHT not to exceed 30 SECONDS
APPROVED MANEUVERS: aileron roll (under 360°), barrel roll, chandelle, Immelmann, loop, wingover

- Slats - LOCK/UNLOCK "LSHIFT+LALT+S" for formation flying (ground crew must perform this)

Engine Start Checklist

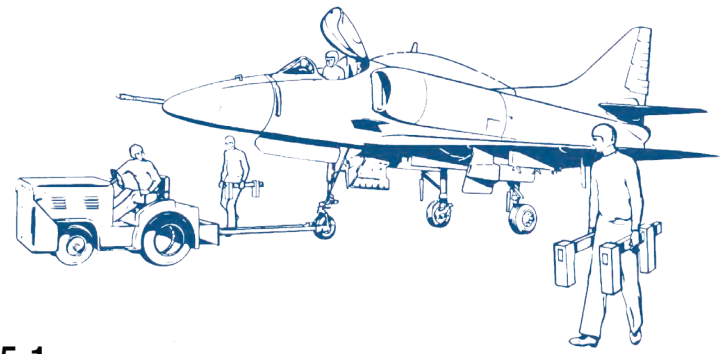
1. Canopy Lever. . . . . OPEN - 6-3, 14
2. Ground Power . . . . . .ON
3. Engine Starter Switch . . . . . DEPRESS - 6-3, 8.
4. Tachometer . . . . . Check RPM - 6-2, 21.
   - A. 5% RPM . . . . . THROTTLE STEP to IGN - 6-1, 4.
   - B. 15% RPM. . . . . THROTTLE STEP to IDLE
   - C. 40% RPM . . . . . Ground Power OFF
   - D. 55% RPM . . . . . Check ENGINE STABLE
5. Caution Panel . . . . . Check FUEL BOOST OFF - 6-2, 6.
6. Oil Pressure . . . . . 35 - 50 PSI - 6-2, 18.
7. Exhaust Temperature . . . . . Check 200 - 340° C - 6-2, 22.

Keep canopy OPEN until ground power is off, and all refueling and rearming procedures are complete.

Equipment Warm-Up Times

Systems and that require warm-up recieve generator power once the engine is stable at 55% RPM.

AN/APG-53A Radar . . . . . 3 MIN. - 6-3, 10.
AFCS . . . . . 90 SEC. - 6-3, 12.
AN/APN-153 Doppler Navigation Radar . . . . 5 MIN. - 6-4, 2.
AN/ASN-41 Navigation Computer . . . . . 2 MIN. - 6-4, 3.
AN/ARC-51A UHF Radio . . . . . 3 MIN. - 6-4, 5.
AN/ARN-52 TACAN . . . . . 3 MIN. - 6-4, 7.
AN/ARA-63 MCL (ILCS) . . . . . zero (solid state) - 6-4, 11.



Pre-Taxi Checklist

1. Ground Power . . . . . OFF
2. Canopy Lever. . . . . CLOSED - 6-3, 14
3. AN/ARC-51A UHF Radio Mode . . . . . T/R+G - 6-4, 5.
4. AN/ARN-52 TACAN Mode . . . . . REC - 6-4, 7.
5. AFCS Standby Switch . . . . . STBY - 6-3, 12.
6. Altimeter . . . . . Set PRESSURE - 6-2, 7.
7. Master Test Switch . . . . . ON, - 6-2, 29. (Hold)
   Check CAUTION PANEL & DASH WARNING - 6-2, 6. & 16.
   A. Liquid Oxygen Indicator . . . . . Check - 6-2, 28.
   B. Fuel Quantity Indicator . . . . . Check - 4-4
8. AN/APN-153 Doppler Nav Radar Mode . . . . TEST - 6-4, 2.
   A. Check SPEED reading . . . . . 121 ± 5 KIAS
   B. Check ANGLE reading . . . . . 0 ± 2°
9. AN/ASN-41 Navigation Computer - 6-4, 3.
   A. AN/ASN-41 Function Selector . . . . . TEST
      A1. BDHI switch . . . . . NAV CMPTR - 6-2, 29.
      A2. Check WIND SPEED . . . . . 223 KIAS - 6-4, 3.
      A3. Check HEADING . . . . . 91° - 6-4, 2.
      A4. Check BDHI POINTER 2 . . CURRENT LOC.
      A5. Check BDHI POINTER 1 . . . . . 30° RIGHT
   B. AN/ASN-41 Function Selector . . . . . STBY - 6-4, 3.
      B1. Input . . . . . MAGVAR, WIND DIRECTION, WIND SPEED, and WAYPOINTS
10. Oxygen Switch . . . . . ON - 6-3, 15.
11. AN/APG-53A Radar Mode . . . . . STBY - 6-3, 10.
12. Wheels and Flaps . . . . Check WHEELS DOWN - 6-3, 1.
13. Master Exterior Lights Switch . . . . . ON - 6-1, 1.
    A. Day CVN . . . . . EXTERIOR LIGHTS OFF - 6-3, 6.
    B. Night CVN . . . . . EXTERIOR LIGHTS ON Excepting TAXI LIGHT
    C. Day/Night AF . . . . . EXTERIOR LIGHTS ON
14. Exhaust Temperature Indicator . . . 200 - 340° C - 6-2, 22.

- Ground handling, 5-4

Pre-Takeoff Checklist

1. AN/APN-141 Radar Altimeter . . . . . ON, Set LAWS INDEXER - 6-2, 26.
2. Armament Panel - 6-2, 34.
   - A. Master Armament Switch . . . . . OFF - 3-1, 1.
   - B. Gun Charging Switch . . . . . SAFE - 3-1, 2.
   - C. Station Select Switches . . . . . DOWN / OFF - 3-1, 3.
   - D. Bomb Arm Switch . . . . . OFF - 3-1, 4.
   - E. Weapon Function Selector Switch . . . . . OFF - 3-1, 5.
3. AN/APN-153 . LAND/SEA AS REQ'D (Light Extg.) - 6-4, 2.
4. AN/ASN-41 . . . . . Set MODE to STBY, Check MAG VAR, WIND DIRECTION, WIND SPEED, and WAYPOINTS - 6-4, 3.
5. AN/ASN-41 Function Selector Switch D1
6. Oxygen Switch . . . . . ON - 6-3, 15.
7. AFCS Panel - 6-3, 1.
   - A. AFCS Standby Switch . . . . . STBY - 4-1, 1.
   - B. AFCS Stability Augmentation . . . . STAB AUG - 4-1, 3.
8. AN/APG-53A Radar Mode Switch . . . . . STBY - 6-3, 10.
9. Spoiler Arm Switch . . . . . ARM - 6-3, 5.
10. APC Power Switch . . . . . OFF (Light Extg.) - 6-4, 4.
11. Chaff Power Switch . . . . . ON - 6-2, 11.
12. MCL Power Switch . . . . . ON - 6-4, 11.
13. MCL Power Switch . . . . . Perform BIT TEST - 1-3
14. MCL Channel Selector Knob . . . . . AS REQ'D
15. ECM Panel . . . . . APR-25 Switch ON, Function Selector to STBY (Light Extg.), set PRF & MSL VOLUME - 6-2, 1. - 3-7
16. Rudder Trim Switch . . . . . Center - 6-1, 5.
17. Trim Position Indicator Panel . . . . . Rudder: 0 Nose: 8 UP - 6-4, 1.
18. Flaps . . . . . ½ to FULL - 6-3, 1.

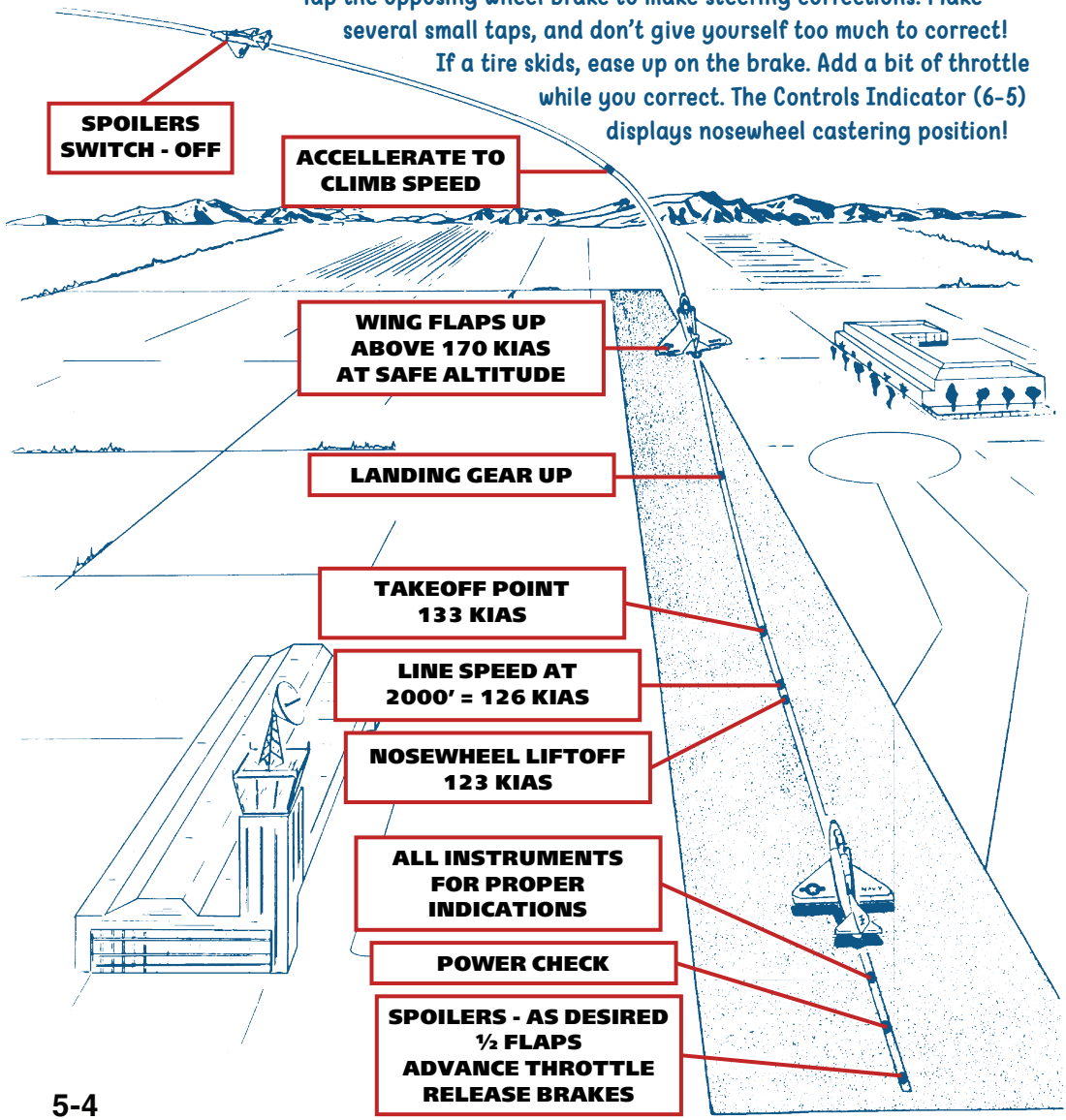
- Takeoff diagram, 5-4

Taxi

1. Extend flaps, open speedbrake & actuate spoilers - 6-1
2. Wheels roll at 75% RPM - 6-2, 21.
3. Reduce throttle to 65-70% RPM
4. Pump brake in direction of turn to caster nosewheel - Happy feet!
5. Thrust overcomes wheel brakes at 90% RPM - 4-5
6. Takeoff point speed increases with weight (up to 160 KIAS at maximum load)

Takeoff

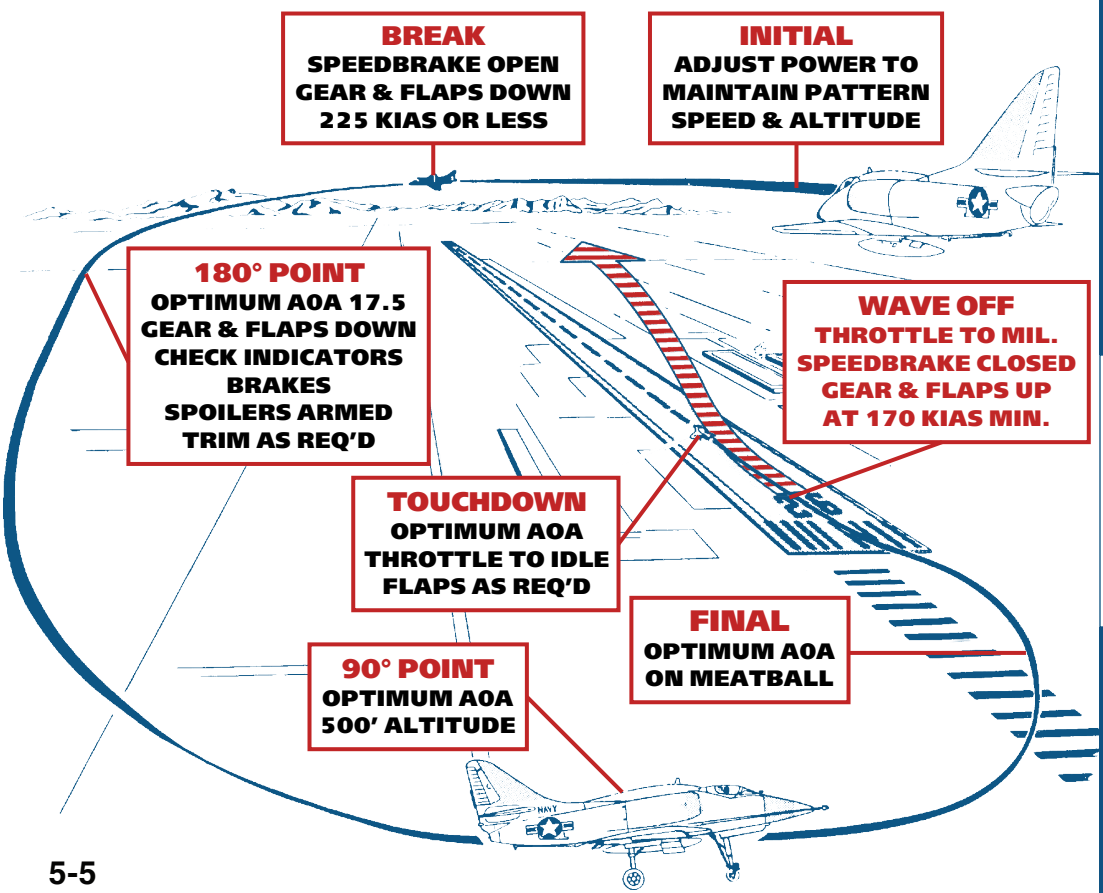
Apply wheelbrakes brake toward the turn. Keep a low speed, but don't stop. Tap the opposing wheel brake to make steering corrections. Make several small taps, and don't give yourself too much to correct! If a tire skids, ease up on the brake. Add a bit of throttle while you correct. The Controls Indicator (6-5) displays nosewheel castering position!



Landing Checklist

1. Break . . . . . Check AIRSPEED 225 KIAS - 6-2, 9.
2. Speedbrake Switch . . . . . OPEN - 6-1, 3.
3. Flaps . . . . . ½ to FULL - 6-3, 1.
4. Landing Gear Handle . . . . . DOWN - 6-2, 5.
5. Spoiler Arm Switch (AF only) . . . . . ON - 6-3, 5.
6. Arresting Hook Handle (CVN only) . . . . . DOWN - 6-2, 23.
7. 90° Point . . . . . Check AIRSPEED 170 KIAS, - 6-2, 9.
   Check ALTITUDE 500' - 6-2, 7.
8. Touchdown . . . . . Check AIRSPEED 125 KIAS - 6-2, 9.
9. Throttle Position . . . . . 0 / IDLE - 6-1, 4.
10. Stick forward, deflect into wind, wings level
11. Rudder . . . . . AS REQ'D
12. Depress wheelbrakes as rudder loses efficacy

- Weight, glide slope maintenance and minimum rate of descent, 4-5
- CVN landing procedures, 3-2 (Case I & APC), 4-3 (Case III)



Engine Flame Out Checklist

1. Throttle Position . . . . . 0 / IDLE - 6-1, 4.
2. Throttle Step . . . . . OFF
3. Emergency Generator Release Handle . . . PULL - 6-2, 33.
4. Check for Fire - 6-2, 16.
   - A. Fire . . . . . EJECT
   - B. No Fire . . . . . RESTART

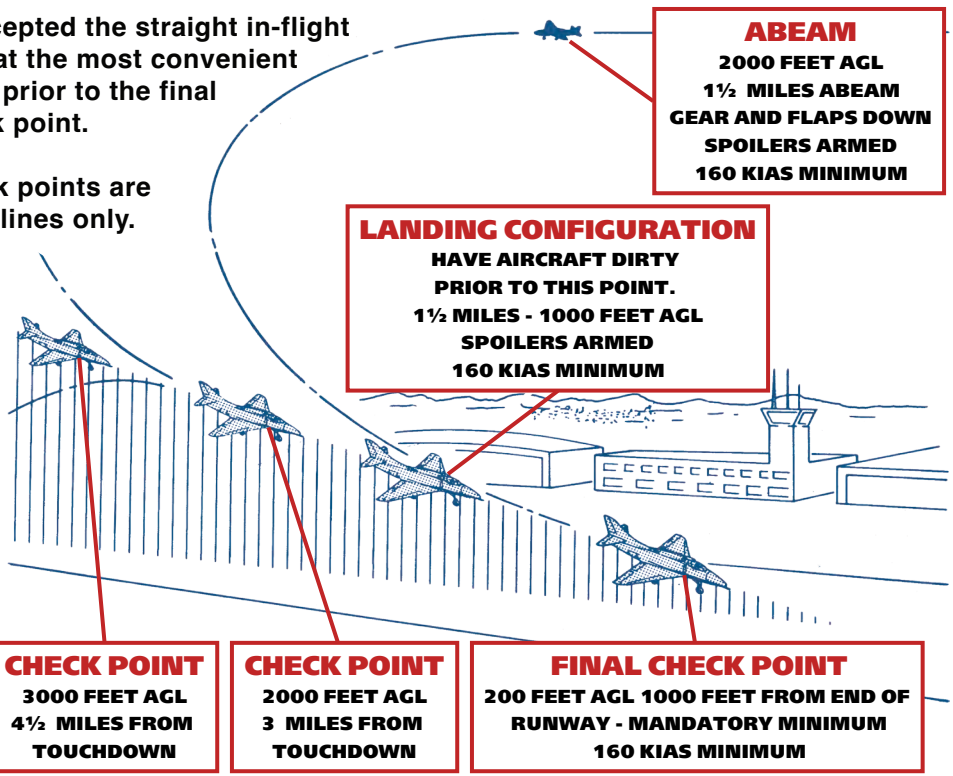
Air Start Checklist - If under 1500' AGL, eject! "LCTRL+E" (3 times)

1. Altitude . . . . . Check above 5000' - 6-2, 7.
2. Throttle Step . . . . . OFF - 6-1, 4.
3. Emergency Generator . . . . . DEPLOYED - 6-2, 33.
4. Fuel Control Switch . . . . . MANUAL - 6-3, 8.
5. Establish Glide . . . . . 250 KIAS - 6-2, 9.
6. Throttle Position Lock Step . . . . IGN then IDLE - 6-1, 4.
7. Throttle Position . . . . . AS REQ'D

Low Precautionary Approach

Intercepted the straight in-flight path at the most convenient point prior to the final check point.

Check points are guidelines only.



High Precautionary Flameout Approach

AIRSPEED & ALTITUDE	OPTIMUM	USABLE BAND
Approach speed	180 KIAS	170 to 200 KIAS
Initial point altitude	7000’	6000’ (gear up)
180° point altitude	4500’	4000’ to 5000’ (Gear Down)
90° point altitude	3000’	2800’ to 3200’ (Gear Down)
Final approach altitude	1000’	500’ minimum
Flare	200’	200’ to 300’

OPTIMUM CONFIGURATION:
Gear down, flaps up.
Altitudes shown are terrain clearance.

DESCENT ENTRY
CLEAN GLIDE, USE S-TURNS AND/OR GEAR EXENTION TO ARRIVE OVER INITIAL POINT AT 7000 FEET AGL

INITIAL TURN
MAINTAIN 20° TO 30° BANK TURN AT 180 KIAS, DELAY INITIAL TURN VARY BANK ANGLE OR LEAVE GEAR UP, IF NECCESARY, TO ARRIVE AT 180° POINT AT OPTIMUM ALTITUDE

INITIAL POINT
7000 FEET AGL, OVER TOUCHDOWN POINT, 90° RELATIVE TO RUNWAY HEADING
180 KIAS, GEAR DOWN, FLAPS UP
START 20° TO 30° GLIDING TURN
RATE OF DESCENT AT 4000 FPM

For precautionary and practice approaches,simulate flameout conditions by setting 75% power, with speedbrakes extended.

This pattern assumes aircraft gross weight is14,000 lbs. (empty external tanks, no stores). If fuel is exhausted, speed and altitude may be slightly lower. Add 5 KIAS for every 1000 lbs. above 14,000 lbs. gross weight.

180° POINT
4500 FEET AGL, 1½ MILES ABEAM LANDING POINT. GEAR DOWN, FLAPS UP, 180 KIAS, SPOILERS ARMED

90° POINT
3000 FEET AGL,
5000 FEET DOWNWIND FROM TOUCHDOWN POINT.
GEAR DOWN, FLAPS UP,
180 KIAS

TOUCHDOWN
140 TO 150 KIAS

FLARE
200 FEET AGL
180 KIAS

FINAL
1000 FEET AGL
180 TO 200 KIAS

Throttle Panel - 6-3, 7.

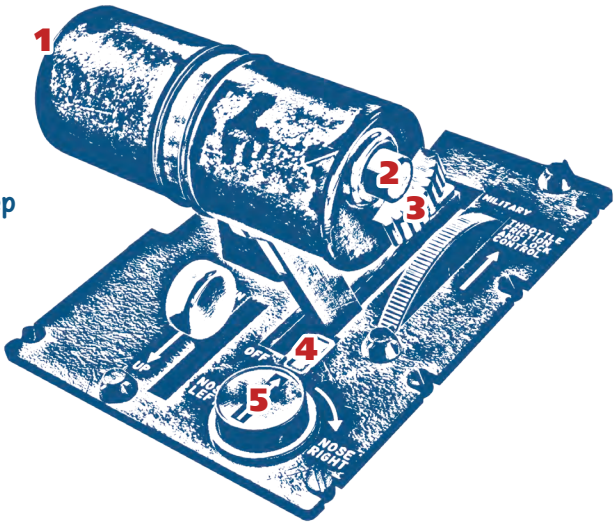
1. Master Exterior Lights Switch - "L"
- Switch located on outside of throttle grip
- Exterior Lights Control Panel, 6-4, 6.

2. Radio Microphone Push to Talk (PTT) - "RALT+\"
- AN/ARC-51A UHF Radio, 1-2

3. Speedbrake Switch - "B"
- Caution Panel, 6-2, 6.

4. Throttle Position Lock
- Zero throttle input to adjust step to or from OFF/IGN/HOME
- Step Up - "RSHIFT+HOME"
- Step Down - "RSHFIT+END"

5. Rudder Trim Switch
- "RALT+Z, RALT+X"



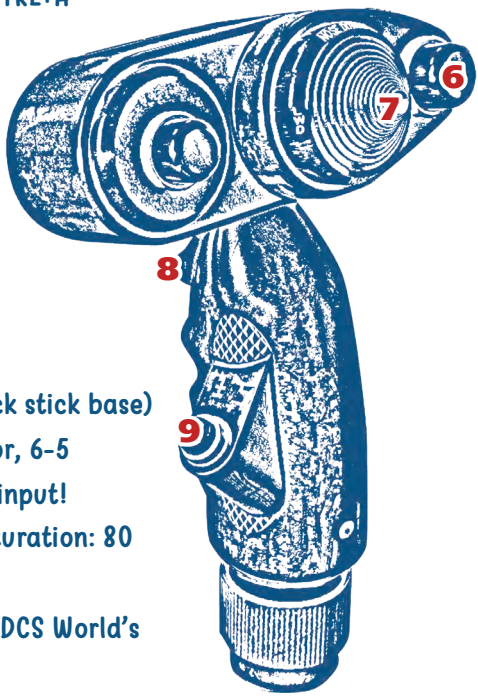
Control Stick

6. Autopilot Override (AFCS Emergency Disconnect) - "LCTRL+A"
- Autopilot, 4-1

7. Trim Switch
- Trim Position Indicator Panel, 6-4, 1.

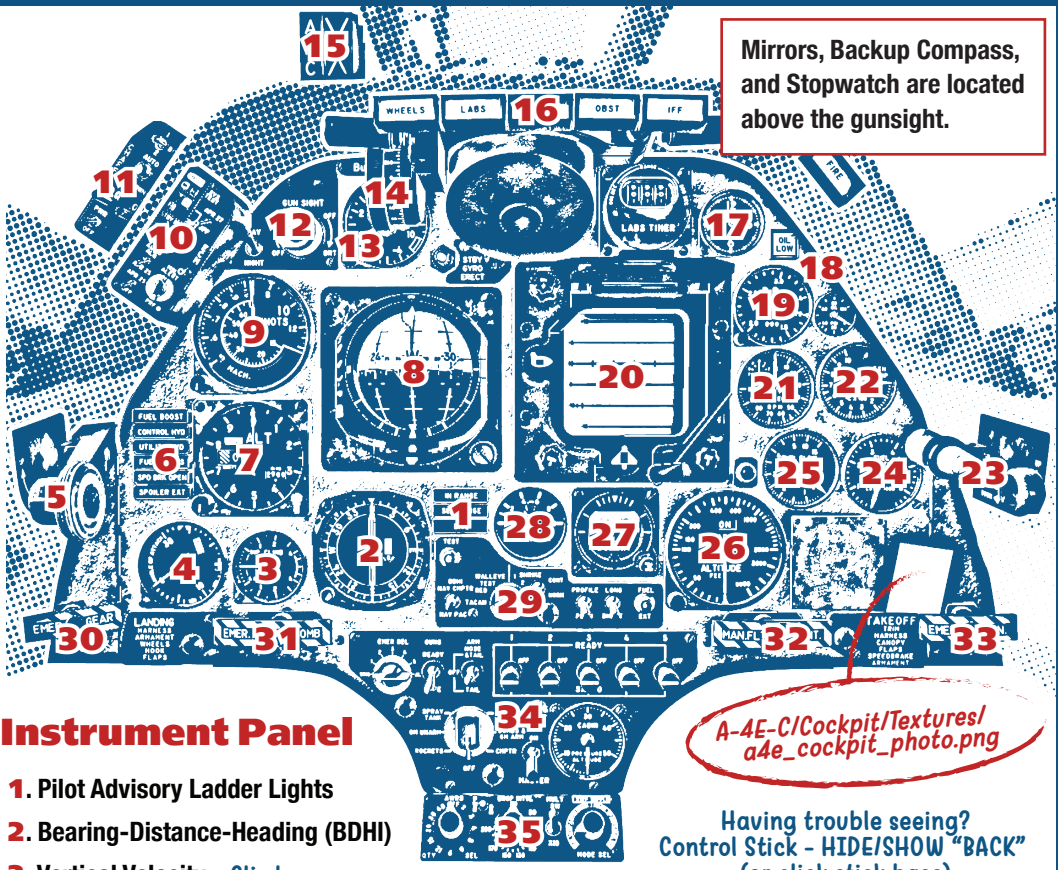
8. Gun-Rocket Trigger - "SPACE"
- Weapons, 3-1, 3-3, 3-4, 3-5

9. Bomb Release Button - "LALT+SPACE"
- Weapons, 3-1, 3-2, 3-3, 3-5, 3-6



- Control Stick - HIDE/SHOW - "BACK" (or click stick base)
- GROUND HANDLING, 5-4 - Controls Indicator, 6-5
- ROLL TOO SENSITIVE? - Adjust the roll axis input!
  - Adjust roll axis input for Curves: 20, Y saturation: 80

THROTTLE OR TRIM TOO SENSITIVE?
- Additional input options are available in DCS World's Special options menu for the A-4E-C!



Mirrors, Backup Compass, and Stopwatch are located above the gunsight.

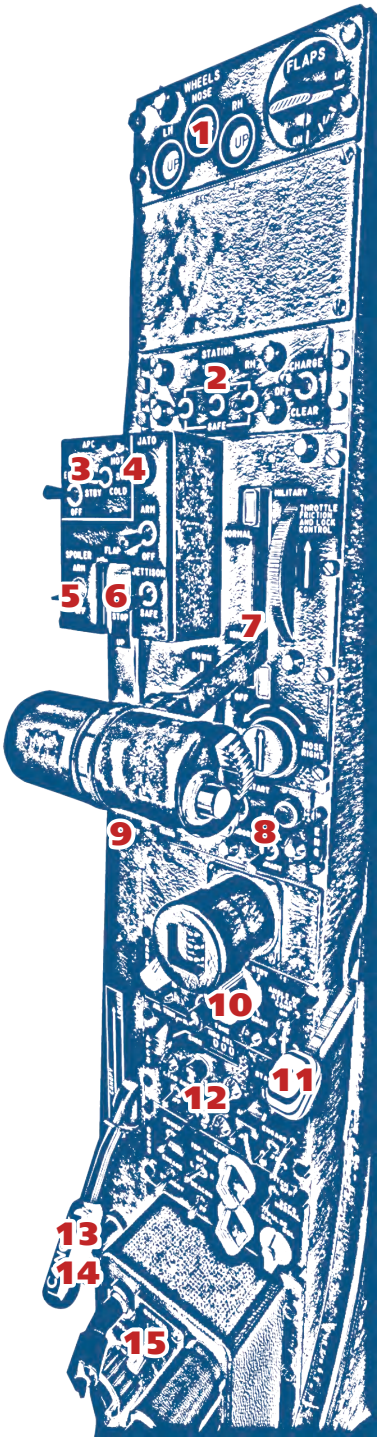
A-4E-C/Cockpit/Textures/a4e_cockpit_photo.png

Having trouble seeing?
Control Stick - HIDE/SHOW "BACK"
(or click stick base)

Instrument Panel

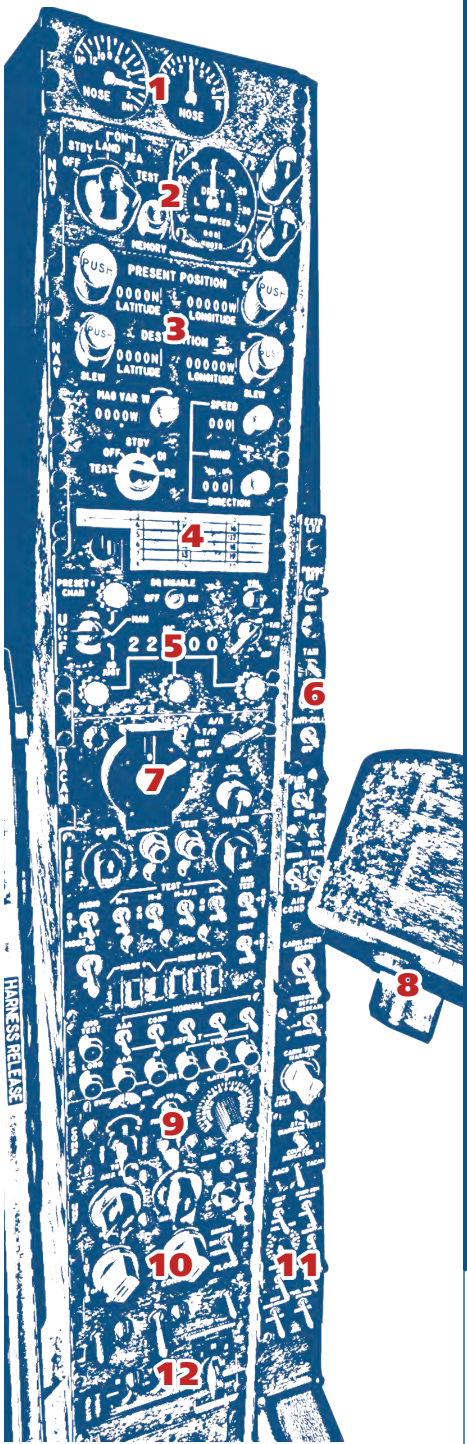
1. Pilot Advisory Ladder Lights
2. Bearing-Distance-Heading (BDHI)
3. Vertical Velocity - Climb
4. Angle-of-Attack - Units
5. Landing Gear Handle - "G", 5-4
6. Caution Panel Ladder Lights
7. Altimeter - Feet
8. All-Attitude (ADI) - MCL V/H, 1-3
9. Airspeed - Knots
10. ECM Panel - 3-7
11. Chaff Panel - 1-4, 3-7
12. Gunsight Panel
13. Accelerometer - 4-5
14. Gunsight Elevation - "RCTRL+[, RCTRL+]"
15. APC & Angle-of-Attack Indexer - Dimmer
16. Dashboard Warning Indicator Lights
17. 8-Day Clock
18. Oil Pressure & Switch
19. Presure Ratio
20. AN/APG-53A Radar Indicator - 2-2
21. Tachometer
22. Exhaust Temperature
23. Arresting Hook Handle - "LALT+G", 4-2, 4-3
24. Fuel Quantity - 4-4
25. Fuel Flow - 4-4
26. AN/APN-141 Radar Altimeter - 2-1
27. Standby Attitude
28. Liquid Oxygen - 6-3, 15.
29. Miscellaneous Switches Panel
30. Emergency Landing Gear Release Handle
31. Emergency Stores Release Handle - 3-1
32. Manual Flight Control Handle
33. Emergency Generator Release Handle - 5-6
34. Armament Panel - 3-1
35. Aircraft Weapons Release System Panel - 3-2

Left Console Control Panels

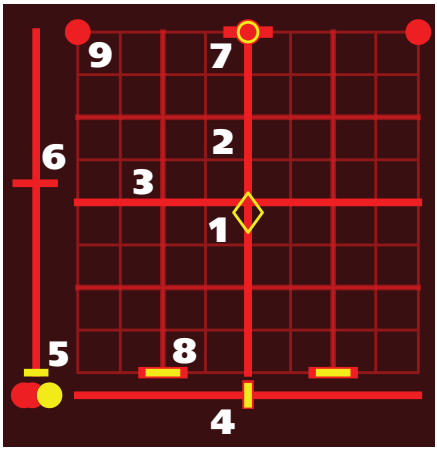


1. Wheels and Flaps Position Indicator Panel
   - Landing Gear Handle, 6-2, 5.
2. Gunpod Panel - 3-1, 3-4
3. APC Control Panel - 4-2
4. JATO Firing Button - "Q"
   - Chaff, 1-4
5. Spoiler Arm Switch - "S"
   - Caution Panel Ladder Lights, 6-2, 1.
   - Actuate: weight on wheels, engine RPM < 70%
6. Flap Handle - "F"
   - When flaps extended, WHEELS dash light blinks
7. Throttle Panel - 6-1
8. Engine Control Panel
   - Engine Start Checklist, 5-1
   - Air Refueling, 4-4
9. Manual Fuel Shutoff Control Lever
10. AN/APG-53A Radar Control Panel - 2-2
    - Additional controls, 6-2, 20. & 29.
11. Shoulder Harness Control Handle
12. Autopilot Control Panel (AFCS) - 4-1
13. Canopy Lever - "LCTRL+C"
14. Emergency Speedbrake Knob
15. Oxygen Switch - "O"

Right Console Control Panels



1. Trim Position Indicator Panel
   - Trim Switch, 6-1, 7.
2. AN/APN-153 Doppler Navigation Radar Control Panel - 1-1
3. AN/ASN-41 Navigation Computer Control Panel - 1-1
4. Radio Preset Channels - 1-2
5. AN/ARC-51A UHF Radio Control Panel - 1-2
   - Radio Microphone Push to Talk, 6-1, 2.
6. Exterior Lights Panel
   - Master Exterior Lights Switch, 6-1, 1.
7. AN/ARN-52 TACAN Control Panel - 1-2
8. White Floodlights Control
9. Compass Control Panel
   - This system not modeled yet
10. Interior Lights Panel
11. AN/ARA-63 MCL (ICLS) Control Panel - 1-3
    - Vert/Horiz guides on ADI, 6-2, 8.
12. AFCS Test Panel
    - This system not modeled yet



- Show controls indicator "RCTRL+ENTER"

The controls indicator is a useful tool for testing bindings for important aircraft controls, and for learning start-up and taxi procedures.

Seach the inputs using the names and terms you see in this manual for the ideal input method for your control setup.

DCS World's Special options menu for the A-4E-C includes trim speed, throttle control, cockpit shake, FFB stick options, and a Simple Braking (Rudder-Assisted) option.

1. *PITCH & ROLL* - Trim Switch, 6-1. 7. - Indicator, 6-4, 1.

Position includes input, curves, trim, and nose-down from speedbrake.

2. *SPEEDBRAKE* - Speedbrake Switch, 6-1, 3. - Emergency Speedbrake Knob, 6-3, 14.

Speedbrakes transit to sides when open. - Caution Panel, 6-2, 6.

3. *FLAPS & SPOILERS*

Flaps transit down when extended. - Flap Handle, 6-3, 6. - Indicator, 6-3, 1.

Spoilers transit up when actuated. - Spoiler Arm, 6-3, 5. - Caution Panel, 6-2, 6.

4. *RUDDER* - Rudder Trim Switch, 6-1. 5.

Enable AFCS Stablity Augmentation Switch for yaw dampener. - 4-1

5. *THROTTLE & THROTTLE POSITION LOCK*

If a "red ghost" throttle input is seen, the axis input is working.

Zero the throttle axis input to adjust Throttle Position Lock step. - 6-1, 4.

6. *ENGINE RPM* - Engine Thrust, 4-4

Useful to learn engine start-up timing. - Start-up, 5-1

7. *NOSE GEAR & NOSEWHEEL CASTER* - Landing Gear Handle, 6-2, 4.

Nose gear pod transits toward center when retracted. - Indicator, 6-3, 1.

Caster nosewheel using differential braking. - Taxi, 5-4

The nosewheel caster indicator's scale is logarithmic. - Responsive at center!

8. *REAR GEAR & WHEELBRAKES* - "W" (Both), "LCTRL+W" (Left), "LALT+W" (Right)

Rear gear pods transit to center when retracted.

Wheelbrakes transit upward when depressed.

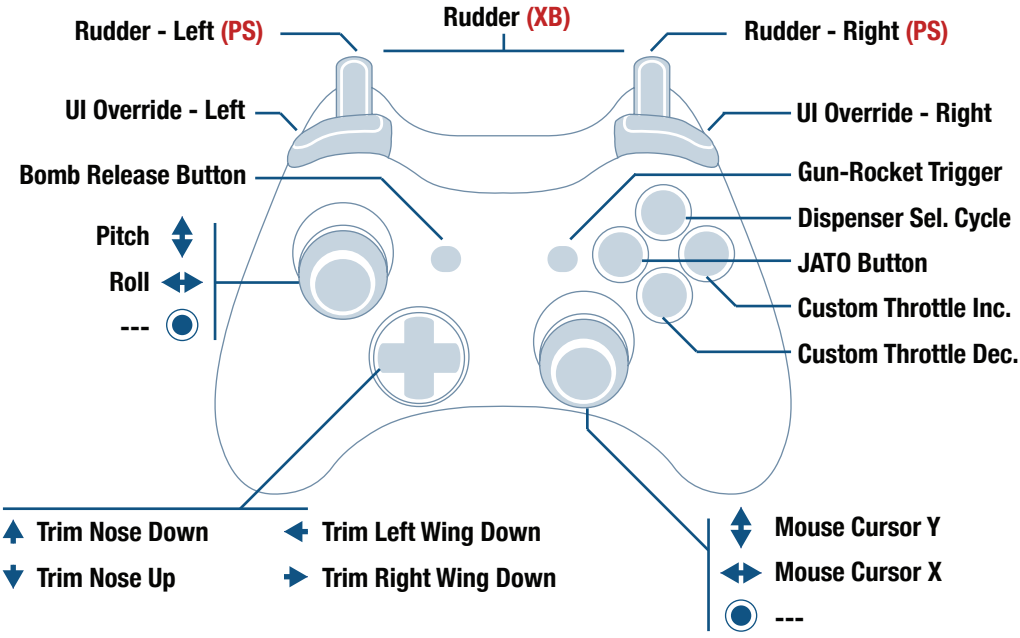
9. *AERODYNAMIC SLATS*

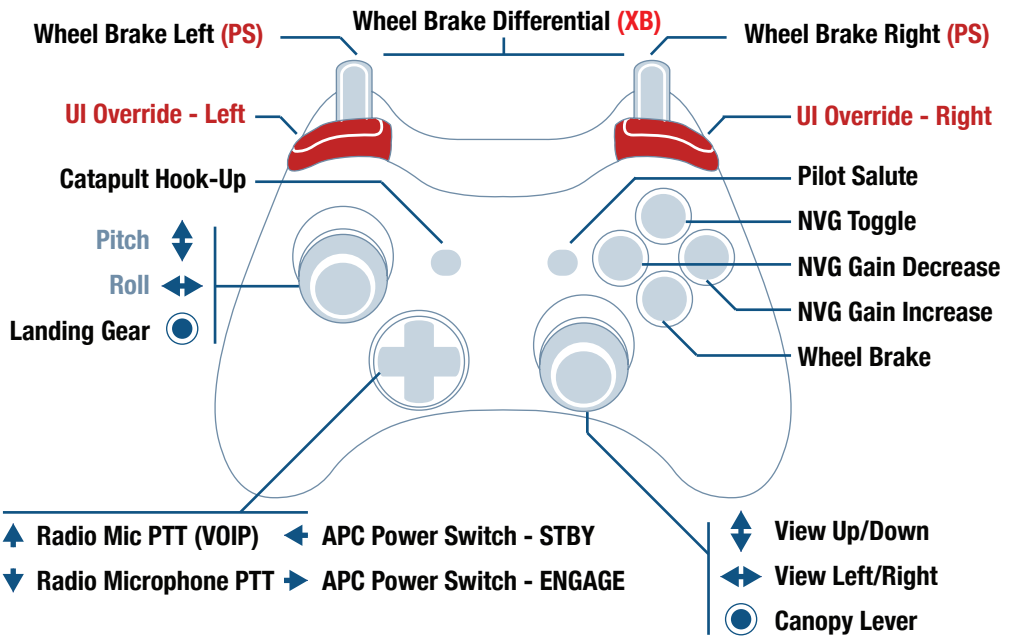
Slat indicators transit downward when collapsed. - Slats Lock, 4-5

UI Override Modifier Buttons

These two inputs increase the functionality of your input devices:
- UI Override - Left inputs LShift + Mouse 2/Right Click.
- UI Override - Right inputs LCtrl + Mouse 1/Left Click.

Using these as modifiers, you can issue an alternate command with the same input. Inputs wthout modifiers fall back to the default layout:





Accessibility and Customizing Your Inputs

Presets for Xbox (XB) and Playstation (PS) controllers are provided in the module's \A-4E-C\Input\Joystick folder. Be sure to calibrate your controller in Windows, and tune axis dead zones, curves, and saturations. Also, use the A-4E-C entry in the DCS Special Menu to set Trim Speeds, Throttle Increment, and Custom Throttle Rate.

